

UNIVERSITY OF CHINESE ACADEMY OF
SCIENCES

DOCTORAL THESIS

**Measurement of differential and
integrated fiducial cross sections for
Higgs boson production in the
four-lepton decay channel in pp
collisions at $\sqrt{s} = 7, 8$ and 13 TeV**

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*A thesis submitted in fulfillment of the requirements
for the degree of Doctor of Philosophy*

in the

Institute of High Energy Physics
Chinese Academy of Sciences

Declaration of Authorship

I, Muhammad AHMAD, declare that this thesis titled, "Measurement of differential and integrated fiducial cross sections for Higgs boson production in the four-lepton decay channel in pp collisions at $\sqrt{s} = 7, 8$ and 13 TeV " and the work presented in it are my own. I confirm that:

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UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

Abstract

Institute of High Energy Physics
Chinese Academy of Sciences

Doctor of Philosophy

Measurement of differential and integrated fiducial cross sections for Higgs boson production in the four-lepton decay channel in pp collisions at $\sqrt{s} = 7, 8$ and 13 TeV

by Muhammad AHMAD

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List of Abbreviations

LAH List Abbreviations Here
WSF What (it) Stands For

Physical Constants

Speed of Light $c_0 = 2.997\,924\,58 \times 10^8 \text{ m s}^{-1}$ (exact)

List of Symbols

a	distance	m
P	power	W (J s ⁻¹)
ω	angular frequency	rad

For/Dedicated to/To my...

Chapter 1

Introduction

The Standard Model (SM) of particle physics describes the nature of the constituents of the matter and the fundamental forces through which they interact. The three fundamental forces (strong, electromagnetic and weak) are described by three dedicated theoretical description Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Electroweak (EWK) theory which is combination of QED and weak are encoded in the Standard Model. In the theoretical framework, the spontaneous symmetry breaking is the primary reason that generated the masses of the particles and separate electroweak (EW) interaction from electromagnetic (EM) interaction [1, 2, 3, 4, 5, 6]. This triggered the search for last missing patch of the SM continued for many years. One of the main motivations of construction of the Large Hadron Collider (LHC) at CERN was also to look for this new particle within full mass range allowed given unitary and above the Large Electron-Positron (LEP) constraints, i.e from 114 GeV to 1 TeV. After a long search, the CMS and ATLAS collaboration, two general purpose experiment at LHC, announced the existence of the heavy scalar boson on 4 July 2012 [7, 8]. Further test to measure the properties of this scalar boson immediately started. The larger dataset allowed to measure the properties of the newly discovered particle. The mass of newly discovered particle, free parameter in theory, is measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [9]. The spin and parity [10] and couplings to other particles [11, 12] has also been measured and found to be compatible with SM predictions. Therefore, the the new particle appear very probable as SM Higgs boson, affiliated with spontaneous symmetry breaking. More precise results are needed to capture any possible deviation from SM prediction which can be doorway to new physics.

Another important property of Higgs boson is to measure its model independent cross section for a particular process, where the cross section illustrate the possibility of the occurrence of that process. This thesis describes the first Higgs boson production cross section with CMS detector at LHC. The measurements are compared with theoretical estimations based on SM. The measurement are made in $H \rightarrow ZZ \rightarrow 4\ell$ decay channel at centre-of-mass energies of $\sqrt{s} = 7, 8, 13$ TeV. All

the cross section are measured in fiducial phase space defined by leptons kinematics and event topology. ATLAS Collaboration has already published similar cross section measurement using same decay channel [13], while measurements in the Higgs to diphoton decay channel have been made by both the ATLAS and CMS collaborations [14, 15]. The Higgs cross section measurement helps to probe the structure, nature of interaction of Higgs boson with decaying particles. The differential measurements are produced as a function of several observables: transverse momentum and rapidity of the four-lepton system, accompanying jet multiplicity, transverse momentum of the leading jet, and difference in rapidity between the Higgs boson candidate and the leading jet using 8 TeV dataset. These differential distributions are sensitive gluon fusion production mechanism, PDFs of colliding proton, theoretical modeling of hard quark radiation and relative contribution of different Higgs boson production mechanism.

The fragmentation of this thesis is systematized as follows. An overview of the SM, Higgs production and decays channel are described in Chapter 2. The overview of LHC and CMS detector design is presented in Chapter 3. The calibration and reconstruction of physics objects in CMS detector is described in Chapter 4. While Chapter 5 presents the measurement of differential and integrated fiducial cross for Higgs boson production in four lepton decay channel. Finally, the conclusions of this thesis are summarized in Chapter 6.

Chapter 2

The Standard Model of Particle Physics

The aim of particle physics is to understand the nature of the constituents of the matter and the fundamental forces through which they interact. There were many theories purposed to understand the laws of particle physics but the Standard Model is the theory which stood firm against high precision tests [16]. The Standard Model is a collaborative theory of the fundamental world, developed in 20th century. The Standard Model express the fundamental particles, boson, fermions and interaction between them by using mediating gauge bosons. Details of fundamental particles and interaction would be described in section 2.1. The last unverified part the Standard Model, the Higgs boson, has also been discovered in 2012 by CMS and ATLAS experiments at LHC at European Organization for Nuclear Research(CERN). The Higgs mechanism, Higgs boson production and decays at LHC would be described in section 2.2 and 2.3. Experimental evidence of the Standard Model would be described in section 2.4. Despite of incredibly successful theory of particle physics, the Standard Model fail to fully describe the known universe. Physics beyond the Standard Model will be described in section 2.5.

2.1 Fundamental Particles and Interactions

2.1.1 Fundamental Particles

Particle with unknown structure is called fundamental particle. Fundamental particles can be divided into two classes, fermions and bosons. Particles which are characterized by Fermi–Dirac statistics and Bose–Einstein statistics are called fermions and boson respectively. According the spin-statistics theorem, particles with integer spin are bosons, while particles with half-integer spin are fermions. Fermions includes all leptons and quarks, and any composite particle made by odd number combination of these, e.g. baryons. Matter is made up of leptons and quarks which

are divided into three “generations” as shown in Table 2.1. All quarks generations include of an “up” type quark with electric charge $+\frac{2}{3}$ and a “down” type with electric charge $-\frac{1}{3}$. Quarks also carry color charge — red, green, or blue, which is responsible for strong interaction. The types of quarks in order of increasing mass are u, d, s, c, b, t. The whole visible matter is only made of lightest type quarks, “up” and “down”. The heavier quarks are only produced in high energy collisions (cosmic rays, particle accelerators) and quickly decays into “up” and “down” quarks.

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. As showed by the theory of the strong force, Quantum Chromodynamics (QCD), a quark antiquark pair or three quarks can combine as mesons ($q_1\bar{q}_2$) or baryons ($q_1q_2q_3$) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of uud and udd quarks respectively.

TABLE 2.1: Generations, charge and interaction types of quarks and leptons.

	Gen. 1	Gen. 2	Gen. 3	Charge	Interactions
Quarks	$\begin{smallmatrix} u \\ d \end{smallmatrix}$	$\begin{smallmatrix} c \\ s \end{smallmatrix}$	$\begin{smallmatrix} t \\ b \end{smallmatrix}$	$\begin{smallmatrix} +\frac{2}{3} \\ -\frac{1}{3} \end{smallmatrix}$	strong, electromagnetic, weak
Leptons	$\begin{smallmatrix} e \\ \nu_e \end{smallmatrix}$	$\begin{smallmatrix} \mu \\ \nu_\mu \end{smallmatrix}$	$\begin{smallmatrix} \tau \\ \nu_\tau \end{smallmatrix}$	$\begin{smallmatrix} -1 \\ 0 \end{smallmatrix}$	strong, electromagnetic, weak

Leptons also have three generations like quarks. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. In fact, quark and gluons are only particles carry color charge. Each generation of leptons have a charged lepton and a neutrino associated. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass $e = 0.511$ MeV, $\mu = 105.7$ MeV and $\tau = 1.78$ GeV. Charged leptons carry +1 and their antiparticle carry -1 electric charge. In addition, all leptons lepton flavor quantum number: all leptons have assigned +1 while antileptons have assigned -1 .

Bosons are the carriers of the fundamental forces, more detailed is described in section 2.1.2. Electromagnetic force is mediated by massless, neutral, spin-1 photon. Weak force is mediated by $W^\pm$ and Z bosons having electric charges ± 1 and 0 respectively. The masses of $W^\pm$ and Z bosons are 80.4 GeV and 91.2 GeV respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge boson which carries color and anti-color charge, thus they can interact with other gluons.

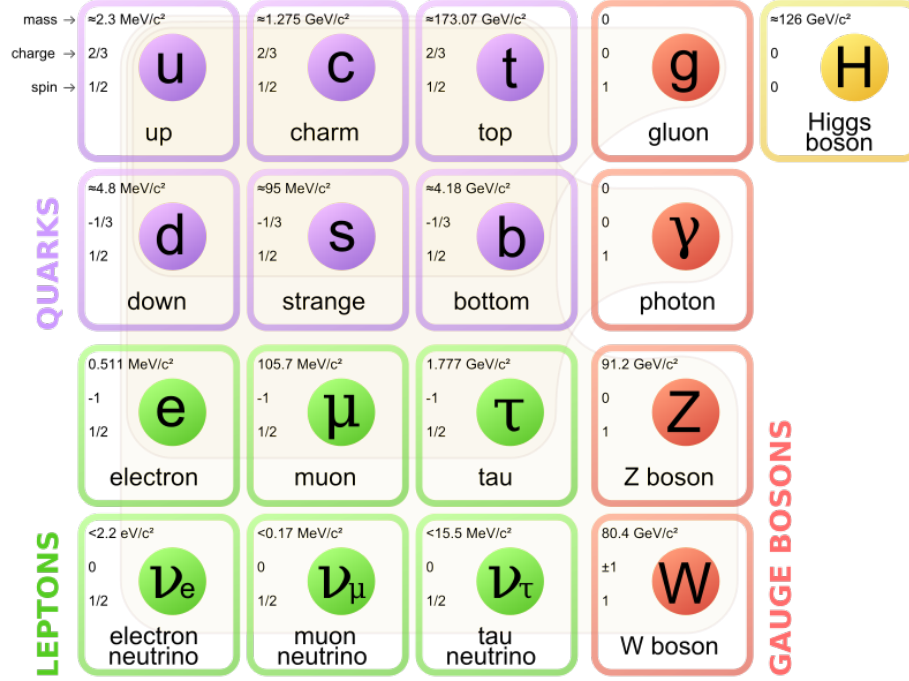


FIGURE 2.1: The Standard Model of elementary particles, with the three generations of matter, gauge bosons in the fourth column, and the Higgs boson in the fifth.

TABLE 2.2: Gauge boson, coupling constants for three forces of the Standard Model

Force	Weak	Electromagnetic	Strong
Exchange Boson	$W^\pm, Z$	photon (γ)	Gluon (g)
Mass(GeV)	80.4, 91.2 GeV	0	0
Coupling Constant	$G_F \approx 1.2 \times 10^{-5} \text{ GeV}^{-2}$	$\alpha = \frac{1}{137}$	$\alpha_s \approx 0.1$
Interaction Range	10^{-16} cm	∞	10^{-13} cm

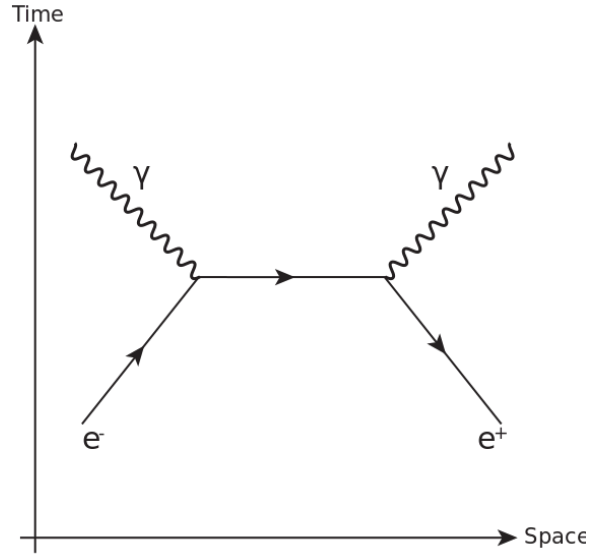


FIGURE 2.2: Feynman diagram of electron positron annihilation.

2.1.2 Fundamental Interactions

The three fundamental forces (strong, electromagnetic and weak) are described by three dedicated theoretical description Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Electroweak (EWK) theory which is combination of QED and weak are encoded in the Standard Model. The Mathematical description of interaction of charged particles are described by QED. QED photon is the force carrier of electromagnetic force. QED is the first theory which explains the interaction between matter and light that patch up special relativity and quantum mechanics. Figure 2.2 shows an example of QED, electron–positron annihilation. Annihilation is the process of destruction of particle and anti particle pair to any set of particles obeying conservation of quantum numbers, energy and momentum. QED is perturbative theory based on fundamentals of quantum mechanics established by P. A. M. Dirac, W. Heisenberg, and W. Pauli during the 1920s. Later, Freeman J. Dyson, Richard P. Feynman and Julian S. Schwinger showed that charged particle can possibly interact through a sequence of processes with growing complexity, and all of these processes could be described by Feynman developed graphical technique. QED is an abelian gauge theory with the symmetry group $U(1)$.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi’s interaction [17]. Weak force is due to exchange of $W^\pm$ and Z bosons. In terms of field

theory, it represented by gauge theory with symmetry group $SU(2)$, having three gauge bosons (W^+ , W^- , Z). Weak interaction rarely called quantum flavordynamics (QFD), it is the only force which is proficient to change the flavor of quarks. Weak force is also held responsible for CP violation. This is very short range force because exchange bosons ($W^\pm$, Z) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, Weak and Electromagnetic forces are unified into Electroweak force above unification energy (on the order of 100 GeV), the Nobel Prize in Physics was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role. In the field theory representation, Electroweak unification is settled under an $SU(2)_L \otimes U(1)_\gamma$ gauge group. Experimental evidence of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino by Gargamelle collaboration in 1973 and the second discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [18, 19, 20, 21].

The Standard Model is represented by $SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma$ gauge symmetry. Where $U(1)_\gamma$ is representation of QED, $SU(2)_L$ of weak theory and $SU(3)_C$ of QCD. By the unification of electromagnetic and weak force represented by $SU(2)_L \otimes U(1)_\gamma$ gauge symmetry [22], Glashow, Salam and Weinberg was able to formulate the fields to account the $W^\pm$, Z and γ (A) boson in footings of the original basis fields A' and B as expressed in equations 2.1, 2.2 and 2.3. The spontaneous symmetry breaking of $SU(2)_L \otimes U(1)_\gamma$ gauge symmetry is the primary reason that generated the masses of the particles and separate electroweak (EW) interaction from electromagnetic (EM) interaction. At higher energies ($> 100 \text{ GeV}$), both EW and EM acts as similar style. More details are explained in section 2.2.

$$W_\mu^\pm = \sqrt{\frac{1}{2}}(A_\mu^1 \pm iA_\mu^2) \quad M_W = \frac{1}{2}g v \quad (2.1)$$

$$Z_\mu^\pm = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \quad M_Z = \frac{1}{2}v \sqrt{g^2 + g'^2} \quad (2.2)$$

$$A_\gamma = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 + g'B_\mu) \quad M_A = 0. \quad (2.3)$$

The strong interaction, one of four fundamental forces, is responsible for strong nuclear force which bounds the nucleus of an atom. As the name speaks for itself, strong force is strongest among four fundamental forces, although it act at a distance of 10^{-15} meter (femtometer) or less. Strong force is described by QCD and only exist between quarks and gluons which carry non-vanishing color charges. In the field theory representation of the Standard Model, the strong interactions is

represented by the $SU(3)_C$ symmetry group. Here, C represents QCD color, which must be conserved in all strong interactions. The Strong force is mediated by gluons. Contrary to the other fundamental forces, strong force does not diminish with gain in distance between quarks known as color confinement. As an implication of this property, Individual free quarks has never observed. No signature of elementary quark or gluon particle are observed in High Energy Physics (HEP) Colliders, instead they bound with each other to form streams of hadrons by phenomenon known as hadronization called jets.

2.2 The Brout-Englert-Higgs Mechanism(BEH)

Gauge fields must acquire mass in order correctly describe weak interaction and to be consistent experimental findings. Before electroweak symmetry breaking (EWSB), all particle supposed to be massless, however a mass term in electroweak Lagrangian for gauge bosons necessary must be introduced to guarantee the renormalization of theory that would violate gauge invariance. Which hints that some basic thing is missing in the theory apart from fermions and gauge boson fundamental fields, which maintain the local gauge invariance. The Higgs mechanism was postulated as a part of Standard Model, which introduces the existence of a new scalar field, so called Higgs field. The Higgs field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of mass of fermions and boson ($W^\pm$, Z boson) are introduced [1, 2, 3, 4, 5, 6]. The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^1 + i\phi^2 \\ \phi^3 + i\phi^4 \end{pmatrix} \quad (2.4)$$

which can be substituted in electroweak lagrangian given below

$$\mathcal{L}_{EWSB} = (D_\mu \phi)^\dagger (D_\mu \phi) + V(\phi^\dagger \phi) \quad (2.5)$$

here $D_\mu = \delta_\mu -igt_a W_\mu^a + \frac{i}{2}g'Y B_\mu$ represents covariant derivative. The lagrangian described in Equation 2.5 is invariant under $SU(2)_L \otimes U(1)_\gamma$ transformation. The kinetic energy term is written in terms of covariant derivatives, and the potential only depends upon on $\phi^\dagger \phi$. The scalar field ϕ can be codify in terms of isospin (I_3), hypercharge (Y) and electric charge (Q) quantum numbers:

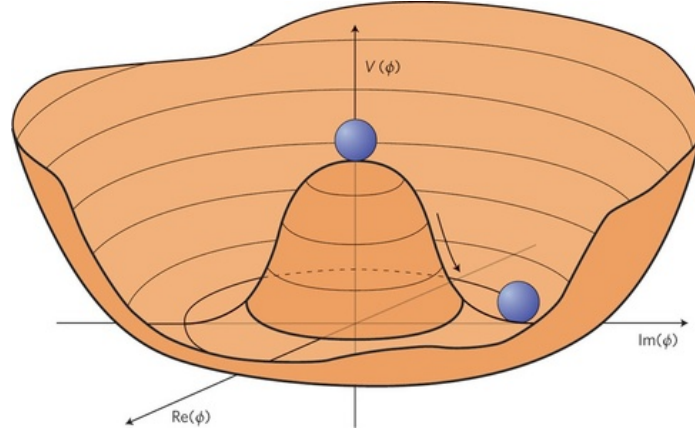


FIGURE 2.3: Shape of the Higgs potential in Eq.2.6

	I_3	Y	Q
$\begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} \\ \frac{1}{-2} \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

while potential term in Equation 2.5 can also be expressed as

$$V(\phi^\dagger \phi) = -\mu^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2 \quad (2.6)$$

here μ is mass term and the parameter λ represents the strength of the interaction. In the scenario, $\mu > 0$ and $\lambda > 0$ and the potential shapes becomes very interesting, essential for Higgs mechanism shown in Figure 2.3, has a minimum for

$$\phi^\dagger \phi = \frac{1}{2}(\phi_1^\dagger + \phi_2^\dagger + \phi_3^\dagger + \phi_4^\dagger) = \frac{-\mu^2}{2\lambda} \equiv \frac{V^2}{2} \quad (2.7)$$

but on the circle of points where the magnitude of z is ϕ . This minimum does not exist at a single value of ϕ , but at circular non-zero values. The $(\phi^\dagger \phi^0)$ selection that affiliated with the ground state (i.e. the lowest energy state, or vacuum) is arbitrary and the selected point is not invariant under rotations in the $(\phi^\dagger \phi^0)$ plane: this is called spontaneous symmetry breaking. If one select to settle the ground state on the ϕ^0 axis, the vacuum expectation value of the ϕ field is

$$\langle \phi \rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ V \end{pmatrix}, \quad V^2 = -\frac{\mu^2}{\lambda}. \quad (2.8)$$

The field ϕ can given below can be derived by rewriting it in a generic gauge, on footings of its vacuum expectation value:

$$\phi = \frac{1}{\sqrt{2}} e^{\frac{i}{\nu} \phi^a t_a} \begin{pmatrix} 0 \\ H + \nu \end{pmatrix}, a = 1, 2, 3 \quad (2.9)$$

here ϕ_a and $\phi_4 = H + \nu$ represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations of the massless vector boson $W^\pm$ and Z , as they are massless and scalar. A unitary gauge transformation is fixed by

$$\phi' = e^{-\frac{i}{\nu} \phi^a t_a} \phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ H + \nu \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

so here we have zero expectation value for the Higgs field.

Now if we rewrite the lagrangian in Eq. 2.5 with field ϕ in unitary gauge, $\mathcal{L}_{EWSB}$ can be disintegrated into three components

$$\mathcal{L}_{EWSB} = \mathcal{L}_H + \mathcal{L}_{HW} + \mathcal{L}_{HZ} \quad (2.10)$$

where three terms is given, assuming approximation $V = \mu^2 H^2 + \cos t$ and ignoring higher order terms:

$$\begin{aligned} \mathcal{L}_H &= \frac{1}{2} \delta_a H \delta^a H + \mu^2 H^2 \\ \mathcal{L}_{HW} &= \frac{1}{4} \nu^2 g^2 W_a W^{\dagger a} + \frac{1}{2} H W_a W^{\dagger a} \end{aligned} \quad (2.11)$$

$$\begin{aligned} &= m_W^2 W_a W^{\dagger a} + g_{HW} H W_a W^{\dagger a} \\ \mathcal{L}_{HZ} &= \frac{1}{8} \nu^2 (g^2 + g'^2) Z_a Z^a + \frac{1}{4} \nu (g^2 + g'^2) H Z_a Z^a \\ &= \frac{1}{2} m_Z^2 Z_a Z^a + \frac{1}{2} g_{HZ} H Z_a Z^a. \end{aligned} \quad (2.12)$$

Mass terms for $W^\pm$ and Z bosons are included in Eqs. 2.11 and 2.12: $W^\pm$ and Z bosons have acquired mass and three additional degree of freedom, related to the longitudinal polarization. Also, three of the four Goldstone bosons have vanished from the Lagrangian $\mathcal{L}_{EWSB}$, thus sustaining the total number of degrees of freedom. A new scalar field has been introduced referred as Higgs field, introduced in the theory. The boson associated with Higgs field is called Higgs boson. The mass of the Higgs boson is linked with a free parameter in the theory λ that depends upon the self-interaction of Higgs field.

Briefly, the BEH mechanism ensures the renormalizability of theory and explain

existence of the massive weak gauge bosons, despite of the fact that the fundamental gauge boson in theory still stays to be massless. In fact, the symmetry still exist in equation so using “hidden” is preferred over broken. According to BEH mechanism, fermions also acquire masses via their interaction with Higgs field. The invariance of the minimum for the Higgs field for $U(1)_\gamma$ group shows the electromagnetic symmetry is not spontaneously broken that keeps photon massless.

2.2.1 Vector Boson Masses and Couplings

Eqs. 2.11 and 2.12 shows the masses of $W^\pm, Z$ bosons depends upon a parameter ν which is vacuum expectation value that defines a mass scale property of EWSB and to electroweak coupling constants:

$$\begin{cases} m_W = \frac{1}{2}\nu g \\ m_Z = \frac{1}{2}\sqrt{g^2 + g'^2} \end{cases} \rightarrow \frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos(\theta_W). \quad (2.13)$$

Also, Higgs boson couplings to vector boson can also be obtained by Eqs. 2.11 and 2.12

$$g_{HW} = \frac{1}{2}\nu g^2 = \frac{2}{\nu}m_W^2 \quad (2.14)$$

$$g_{HZ} = \frac{1}{2}\nu(g^2 + g'^2) = \frac{2}{\nu}m_Z^2. \quad (2.15)$$

Using above Eqs. 2.14 and 2.15 helps to get the ratio of Higgs decay to $W^\pm, Z$ bosons

$$\frac{\mathcal{B}(H \rightarrow W^+W^-)}{\mathcal{B}(H \rightarrow ZZ)} = \left(\frac{g_{HW}}{\frac{1}{2}g_{HZ}} \right)^2 = 4 \left(\frac{m_W^2}{m_Z^2} \right)^4 \approx 2.4. \quad (2.16)$$

At last, the characteristic mass scale for EWSB may be obtained by the relation between the Fermi constant G_F and the ν parameter:

$$\nu = \left(\frac{1}{\sqrt{2}G_F} \right) \approx 246 \text{ GeV}. \quad (2.17)$$

2.2.2 Fermions Masses and Couplings

Fermions couples to Higgs boson field to acquire mass by Higgs mechanics. This can be achieved by introducing a $SU(2)_L \times U(1)_\gamma$ invariant term (called yukawa term) in SM lagrangian that describes the interaction between the fermion fields and the Higgs. As ϕ is an isodoublet and fermions can be disintegrated up into right-handed and left-handed singlet, the yukawa term for each fermion will have

expression given below for leptons:

$$\mathcal{L}_\ell = -G_\ell \bar{l}_\ell \phi l_R + \bar{\ell}_R \phi^\dagger l_\ell. \quad (2.18)$$

As the first component of ϕ is zero, so a mass term will only emerge for the second component of l_ℓ : this exactly repeat the fact that neutrino is massless.

$$\mathcal{L}_\ell = -\frac{G_{H\ell}}{\sqrt{2}} \nu \bar{\ell} \ell - \frac{G_{H\ell}}{\sqrt{2}} H \bar{\ell} \ell. \quad (2.19)$$

For quarks, the down type quarks (d, s, b) behave in similar way as leptons while up type quarks (u, c, t) must couple to charge conjugate of ϕ

$$\phi_c = -i\tau_2 \phi^* = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^3 + i\phi^4 \\ -\phi^1 + i\phi^2 \end{pmatrix} \quad (2.20)$$

in unitary gauge that becomes

$$\phi_c = \frac{1}{\sqrt{2}} \begin{pmatrix} \eta + \nu \\ 0 \end{pmatrix}. \quad (2.21)$$

Therefore, the yukawa Langrangian can be expressed as

$$\mathcal{L} = -G_{H\ell} \bar{L}_L \phi l_R - G_{Hd} \bar{Q}_L \phi d_R - G_{Hu} \bar{Q}_L \phi^c u_R + h.c.. \quad (2.22)$$

The couplings constant and mass of a fermion can be derived from Eq. 2.19, written as

$$m_f = \frac{G_{Hf}}{\sqrt{2}} \nu \quad (2.23)$$

$$g_{Hf} = \sqrt{G_{Hf}} \sqrt{2} = \frac{m_f}{\nu}. \quad (2.24)$$

The fermions mass can not be predicted by the theory as G_{Hf} is free parameters. The interaction of Higgs field with fermions transforms right-handed chirality in a left-handed chirality and vice versa in a way that object acquire mass. The strength of the interaction of Higgs fields with particles is reflected by its mass.

This evaluation is rather interpretative, as it not accumulating the fact that there are three families of quarks and leptons in reality. The different between weak mass eigenstates and physical mass eigenstates makes real analysis complicated. Transformation of weak mass basis to physical basis is essential to deal with physical particles. This is achieved by the usual unitary transformation matrices, Cabibbo–Kobayashi–Maskawa (CKM) and Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrices, for quark and lepton sectors respectively.

2.3 Production and Decay of the Higgs Boson

After electroweak unification, spontaneous symmetry breaking and Higgs mechanism solves all the puzzles with the assumption of existence of the Higgs boson. Higgs boson was considered as the last missing patch of the Standard Model. Higgs boson has been searched at LHC within full mass range allowed given unitary and above LEP constraints, i.e from 114 GeV to 1 TeV. In this section, Higgs boson production and decay modes would be discussed. This will allow us to identify the most promising channels in the perspective of Higgs boson discovery and its properties measurements. Theory does not predict the Higgs mass and coupling of the Higgs with bosons are directly proportional to the square of their masses as shown in Eqs. 2.14 and 2.15. In case of fermions, their couplings with the Higgs is directly proportional to the mass of the fermion as in Eq. 2.24. For this reason, the most promising channels for Higgs searches are its decays to W^+W^- , ZZ and third generation of leptons. Also, photon being massless particle does not couple to Higgs directly but through loop contributions dominated by top quark loop and $W^\pm$ boson.

2.3.1 Higgs Production

The dominant production modes of the Higgs boson in proton-proton collisions at LHC are shown in Fig. 2.4, corresponding production cross sections as a function of the Higgs mass at center of mass energies $\sqrt{s} = 7, 8$ and 13 TeV are shown in Fig. 2.5. A considerable increase in the Higgs production cross section is found as moves towards higher center of mass energies shown in Fig. 2.6 for Higgs mass 125 GeV.

Gluon-gluon fusion

This is most dominant Higgs production mode at LHC in whole mass range and center of mass energies range illustrated in Fig. 2.4 (top left). As LHC and tevatron are hadron colliders, it highly probable that two gluons responsible for binding of hadron collide. The gluons couples to Higgs through the loop of virtual quarks, as coupling to Higgs is directly proportional to the mass of fermions, top quark loop is most dominant. The ggH cross section for Standard Model Higgs at $m_H = 125$ GeV at center of mass energies of 7 TeV, 8 TeV and 13 TeV are approximately 15, 19, 44 pb respectively. As we move towards higher energies the production cross section increases significantly, i.e 27% from 7 TeV to 8 TeV and 227% from 8 TeV to 13 TeV at $m_H = 125$ GeV. The cross section numbers in Table 2.3 are computed at NNLO QCD and NLO EW accuracies.

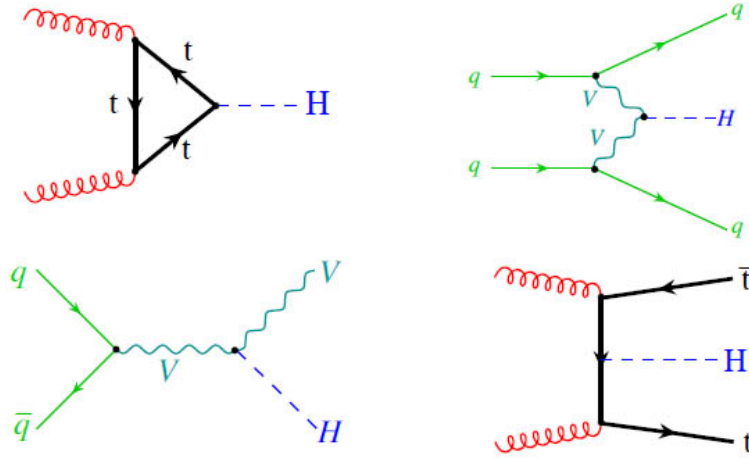


FIGURE 2.4: Higgs production mechanisms at tree level in proton-proton collisions: gluon-gluon fusion (top left); VV fusion (top right); W and Z associated production (bottom left); $t\bar{t}$ associated production (bottom right).

Vector boson fusion

Vector boson fusion (VBF) is second most important Higgs production mode at LHC and LEP as shown in Fig. 2.4 (top right). VBF Higgs production cross section is roughly 10 times lower than ggH at $m_H = 125$ GeV, and becomes comparable at $m_H = 1$ TeV. The VBF cross section for Standard Model Higgs at $m_H = 125$ GeV at center of mass energies of 7 TeV, 8 TeV and 13 TeV are approximately 1.2, 1.6, 3.8 pb respectively. The cross section numbers in the Table 2.3 are computed at NNLL QCD and NLO EW precision. In VBF production mode, two $W^\pm$ or Z boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in forward region of detector. These jets signature helps to increase signal to background ratio.

W and Z associated production

W and Z associated production also called Higgs Strahlung is defined by Higgs production in association with $W^\pm$ or Z boson as shown in Fig. 2.4 (bottom left). In this production mechanism, Higgs is radiated by high energetic $W^\pm$ or Z bosons that is produced by fusion of two fermions. This production mechanism was leading production mode at LEP and sub-leading at Tevatron. At LEP it becomes very

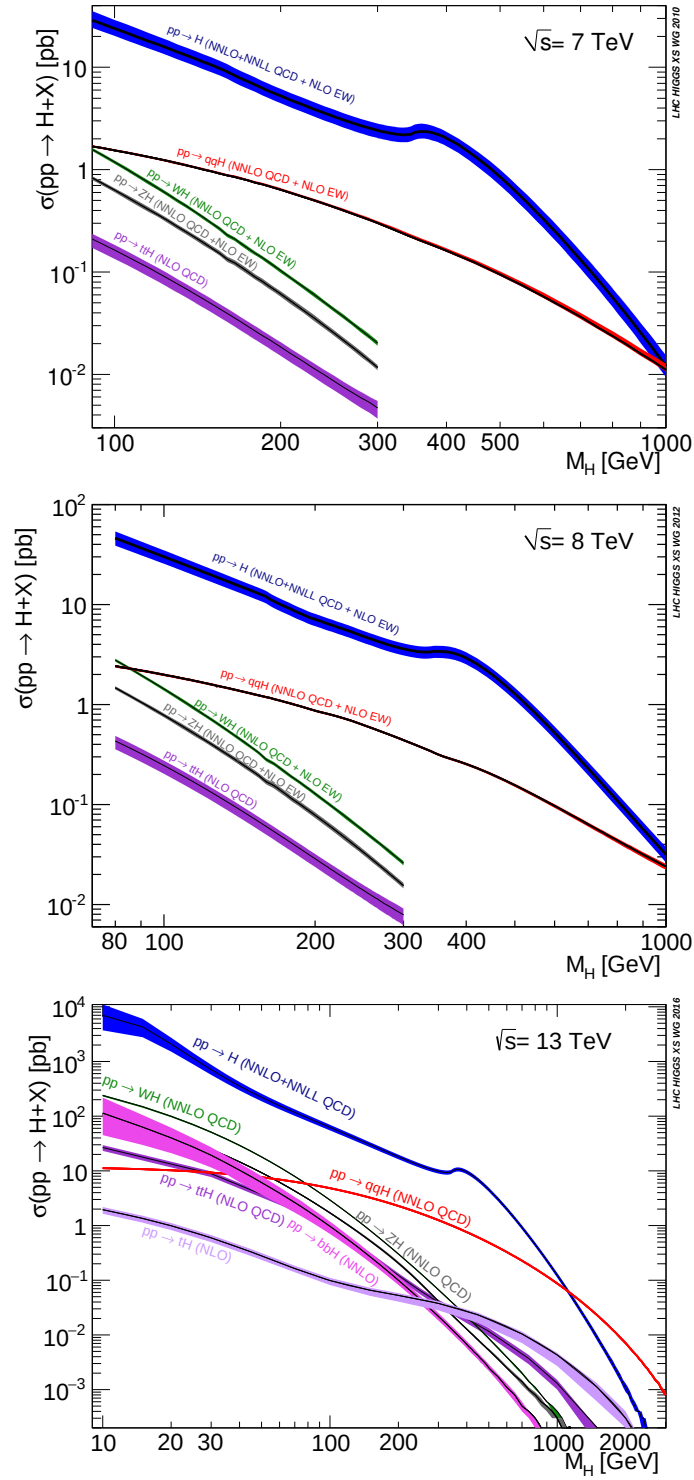


FIGURE 2.5: Standard Model Higgs boson production cross sections at 7, 8 and 13 TeV as function of mass of Higgs [23]

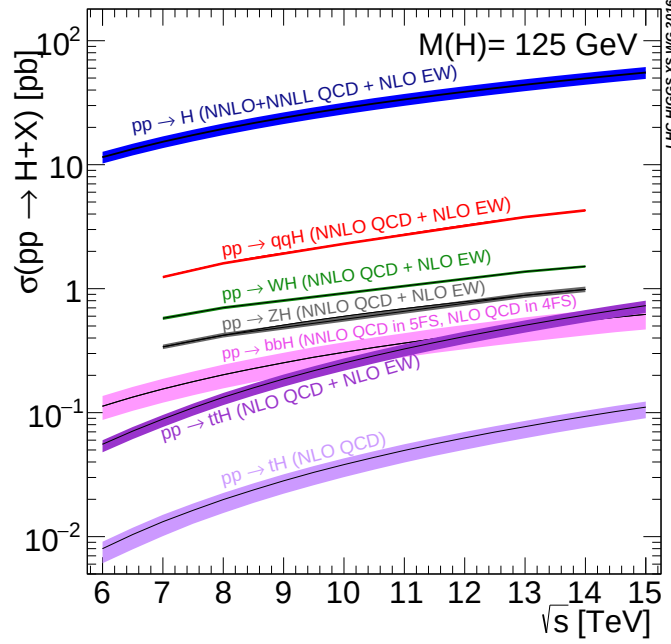


FIGURE 2.6: Higgs production cross sections center of mass energies [23]

TABLE 2.3: Higgs boson production cross section in pb for different Higgs masses by ggH and VBF productions mechanism at LHC

$m_H(\text{GeV})$	ggH			VBF		
	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV
120	16.61	21.07	47.38	1.30	1.68	3.9
125	15.31	19.47	44.14	1.24	1.60	3.78
130	14.15	18.03	41.23	1.19	1.53	3.64

TABLE 2.4: Higgs boson production cross section in pb for different Higgs masses by associative productions mechanism at LHC

$m_H(GeV)$	WH			ZH			ttH		
	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV	7 TeV	8 TeV	13 TeV
120	0.66	0.81	1.57	0.39	0.48	0.99	0.11	0.15	0.57
125	0.58	0.70	1.37	0.34	0.42	0.88	0.09	0.13	0.51
130	0.51	0.62	1.21	0.30	0.37	0.79	0.08	0.12	0.45

probable to produce $W^\pm$ or Z boson with electron positron collisions. In case of hadron collider, tevatron makes quark antiquark collisions more likely as compared to LHC. AT LHC this production mechanism is at third number. The WH and ZH cross section for Standard Model Higgs at center of mass energies of 7 TeV, 8 TeV and 13 TeV are listed in table 2.4 which are computed at NNLL QCD and NLO EW precision.

Top fusion

The least dominant process of Higgs production is illustrated in Fig. 2.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$. The ttH cross section for Standard Model Higgs at center of mass energies of 7 TeV, 8 TeV and 13 TeV are listed in table 2.4 which are computed at NLO QCD and NLO EW precision.

2.3.2 Higgs Decay

As Higgs boson is electroweak particle, so it wants to decay quickly. The Higgs boson decay likelihood depends on a variety of aspects, i.e mass of the Higgs boson, interaction strength etc, most of these factors are controlled by Standard Model except Higgs mass itself. The life time of Standard Model Higgs with mass of 126 GeV is about 1.6×10^{-22} s. Higgs boson can decay to all Standard Model particles with mass but it likes to decay most heavier particles available. Higgs decay branching ratio to a variety of particles are illustrated in Fig. 2.7.

The most dominant decay channel of Higgs boson $H \rightarrow b\bar{b}$ at $m_H \approx 125$ GeV. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, the $b\bar{b}$ channel has been exploited in the boosted regime, in association with a vector boson decaying leptonically, by CMS [24], by ATLAS [25] and by CDF and D0 [26]. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to perfectly reconstruct. Because of this, it

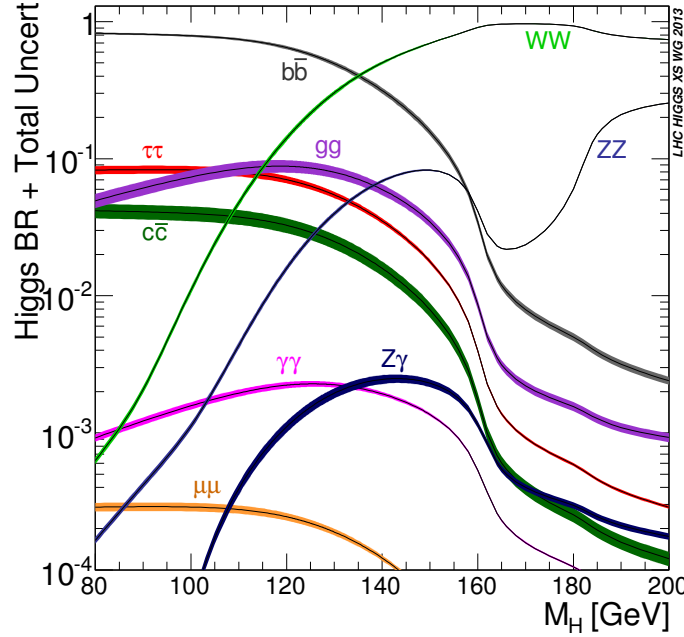


FIGURE 2.7: Branching ratios for different Higgs decay channels as a function of the Higgs mass [23]

is necessary to exploit other channels in this region. $H \rightarrow \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution. The main background in this channel are $q\bar{q} \rightarrow \gamma\gamma$ and converted photon coming from electron conversion after colliding with detector material. As the Higgs mass at 125 GeV, Higgs decay to diboson (WW , ZZ) boson are also available, where one or both boson can be offshell. In case of WW decay, its impossible to completely reconstruct WW boson in detector as it decays $H \rightarrow WW \rightarrow \ell\nu\ell\nu$ have two neutrinos and $H \rightarrow WW \rightarrow \ell\nu jj$ have one neutrino and two jets. These objects pulls mass distribution to be wide and signal to background ratio to be low, which makes WW decay channel less sensitive. While $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can reconstructed with excellence in the detector. The branching ratio of Higgs in intermediate range is illustrated in Fig. 2.8.

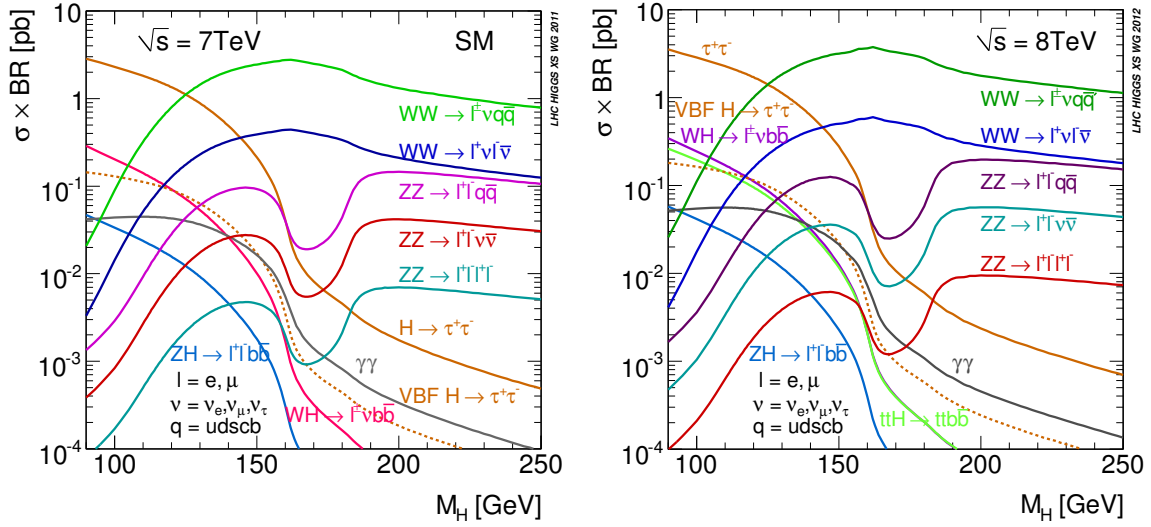


FIGURE 2.8: Higgs cross section times branching ratios 7 TeV (left) and 8 TeV (right) [23].

2.4 Experimental Evidence

The electroweak unification theory lead to series of impressive experimental discoveries. In 1973, the evidence of neutral current through inelastic $\nu_\mu N \rightarrow \bar{\nu}_\mu N$ scattering was found by Gargemelle bubble chamber collaboration at CERN [27]. This evidence provide starting platform to the Standard Model of electroweak interaction.

Super Proton Synchrotron proton-antiproton collider at CERN played its role to next breakthrough that came a decade later. The existence of W, Z boson announced by UA1 and UA2 collaboration through the channels $Z \rightarrow e^+e^-$, $Z \rightarrow \mu^+\mu^-$, $W \rightarrow e\nu$ [18, 19, 20, 21]. This was a first observation of W, Z boson that mediates weak force. Carlo Rubbia and Simon van der Meer was awarded with Nobel Prize for their contribution in 1984.

These important experimental discoveries and growing knowledge on weak interaction convinced to construct the Large Electron Positron collider (LEP) at CERN, for deeper look into W, Z further decays, their precise mass measurement [28, 29]. LEP was equipped with four detectors around the ring; ALEPH (Apparatus for LEP Physics), DELPHI (Detector with Lepton and Hadron Identification), OPAL (Omnipurpose Apparatus for LEP) and L3 (Letter of Intent 3). These experiments reported many precise results of the Standard Model, specially measuring the mass of W and Z boson and confirming three neutrino species consistent with Standard Model [29].

Around 2000, Large Hadron Collider (LHC) [30] at CERN construction started to look for last missing patch of electroweak unification, the Higgs boson. LHC is equipped with four major detectors; ATLAS (A Toroidal LHC Apparatus), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment), LHCb (Large Hadron Collider beauty). LHC recorded data of proton-proton collisions at center of mass energies 7 and 8 TeV in 2011 and 2012. After a long search, the CMS and ATLAS collaboration, two general purpose experiment at LHC, announced the exist of the heavy scalar boson on 4 July 2012 [7, 8]. Further test to measure of the properties of this scalar boson immediately begins, in March 2013, CERN announced the tentative confirmation of consistency of its properties with standard Model [9, 31, 32, 33].

2.5 Beyond Standard Model

Although Standard Model has passed numerous experimental test with great precision, it is not considered to be theory of everything. There are many phenomenons which are not explained by SM i.e Hierarchy problem, existence of dark matter and dark energy, neutrons mass etc. There are several theoretical extension posed in Standard Model such as Minimal Super Symmetrical Model(MSM) and Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical description, such as string theory, extra dimension. The simplest broadening of the SM Higgs sector is the the electroweak singlet (EWS) model [34]. In this model, a heavy singlet is defined along with SM scalar Higgs field and then two CP-even Higgs bosons comes out as result of spontaneous symmetry breaking. Anther generic extension to the SM Higgs is the 2 Higgs Doublet Model (2HDM), which introduces two scalar, complex doublet ϕ_1 and ϕ_2 [35]. Spontaneous symmetry breaking results five Higgs boson, a neutral CP-odd one (A), two CP-even particles (h and H) and two charged ($H^\pm$). Experiment and theory both are looking new any direct or indirect hints of new physics. some of main aspects where Standard Model is unable to explain are discussed below

2.5.1 Hierarchy Problem

The field theory breaks down because of existence of quantum gravity at plank scale ($M_p \approx 10^{19}$ GeV). The larger discrepancy between the features of the gravity and the weak force is known as "Hierarchy Problem". Writing the Standard Model Lagrangian for the Higgs mass will lead to following expression below

$$m_H^2 = m_{oH}^2 + \delta m_H^2 \quad (2.25)$$

here m_{oH}^2 represents mass parameter exist in the Standard Model Lagrangian, and Δm_H^2 represents the quantum radiative corrections to the mass that are predicted to be heavier. These corrections can be found as

$$\Delta m_H^2 \approx \frac{\lambda^2}{16\pi^2} \Lambda^2 \quad (2.26)$$

here λ represents the Yukawa coupling constant, and Λ represents the energy at which the Standard Model is invalid, or about 1 TeV. This was also one of the reason to expect Higgs boson mass at order of few hundred GeV. The Super symmetry (SUSY) provide the solution of Hierarchy problem by predicting the existence of heavier super partners of all SM particles (bosons and fermions). Several direct and indirect searches has been conducted at LHC to hunt for SUSY, but no evidence is found so far.

2.5.2 Dark Matter and Dark Energy

Latest observation from cosmological experiments shows that universe is composed of 4.9% ordinary matter, 26% Dark Matter and 70% Dark Energy. Standard Model does not provide any clue about existence of Dark Matter and Dark Energy.

2.5.3 Matter Antimatter Asymmetry

Today, if we look around in the universe, we see matter dominance over antimatter. At the time of Big-Bang matter and antimatter are supposed to be produced in equal quantities. This give rise to one of the greatest mysteries, where all antimatter has been gone. Standard Model is unable to explain this imbalance between matter and antimatter.

2.5.4 Grand Unification and Beyond

One of main goal of the particle physics, is to unify all fundamental force at some energy scale to describe the theory of everything as Standard Model unify electromagnetic and weak force in to electroweak. But Standard Model does not provide any clue about unification of gravity along with other fundamental force generally referred as Grand Unification Theory (GUT). This implies that Standard Model itself is a patch of Grand Unification. The search for any new physics (beyond Standard Model) is important and justified that could improve our understanding about these unsolved mysteries.

Chapter 3

Experiment

3.1 The Large Hadron Collider

The Large Hadron Collider (LHC), the most powerful particle accelerator of the world that has tendency to smash two protons beams at the energy up to 7.0 TeV per beam. It is the most advanced, complex and largest experiment in the history of mankind, designed and constructed by a collaborative team of more than 10,000 technician, engineers and scientists from 100 countries. The LHC was constructed at CERN from 1998 to 2008, in a tunnel of 27 kilometers(17 mi) circumference below the France-Switzerland border close to Geneva, Switzerland. The LHC is accommodated in a 3.8 meters wide circular tunnel at a depth varies from 75-150 meters underground. Previously, this tunnel also hosted the Large Electron-Positron (LEP) Collider constructed from 1983 to 1988.

The first proton beam was injected into LHC on September 10, 2008. The LHC was able to fire the protons around the tunnel in the segments of three kilometers each time. The initial test to circulate the beams in clockwise and counterclockwise throughout the collider was also successfully completed by the same day. On 19 September 2008, an electric defect triggered the magnet quench in about 100 bending magnets. The consequences of the incident led to an explosive loss of six tonnes of liquid helium that damaged 50 superconducting magnets and their mounts. Three months later, CERN unfolded the details of the incident and published technical report. Particle accelerator was repaired after this incident that took about an year. Finally, on November 20, 2009, LHC became operational again and a low energy beam was successfully rotated within tunnel. On November 30, shortly after, LHC achieved 1.18 TeV per beam and became world's strongest particle accelerator, defeating the Tevatron record of 0.98 TeV. In 2010, LHC kept boasting the beam energies and reached up to 7 TeV center of mass energy (3.5 TeV per beam). In 2011 and 2012, LHC consistently delivered data at 7 and 8 TeV center of mass energies respectively. During 2013 and 2014 , the machine was shutdown to do major parts upgrade on the LHC machine and detectors installed on it. In 2015, upgraded LHC started again at 13 TeV center of mass energy close to full stipulated

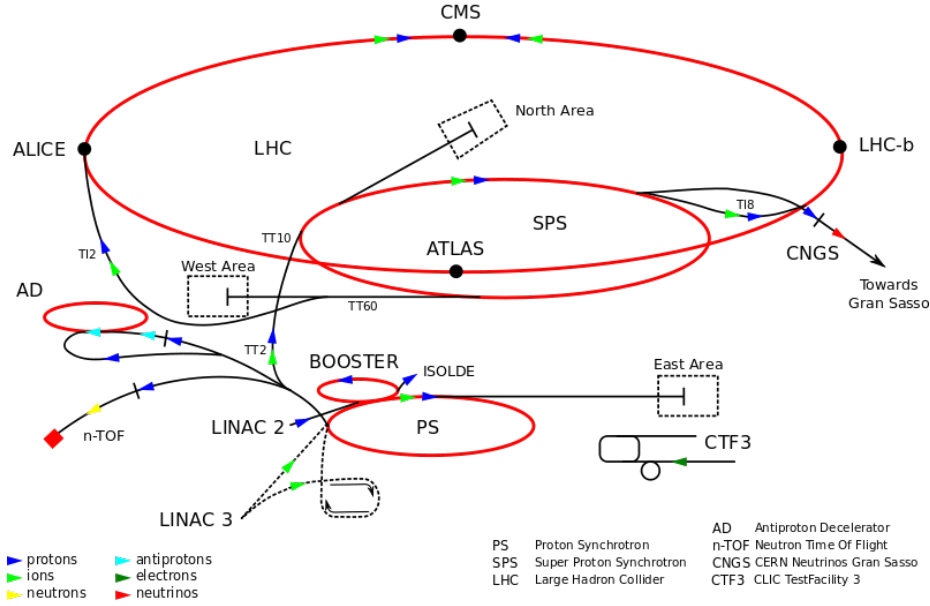


FIGURE 3.1: LHC experiment at CERN

operating energies (14 TeV). Since then, LHC is regularly delivering proton-proton collisions data to explore Higgs and other sectors of particle physics.

3.1.1 Machine Layout and Performance

Machine Layout

The basic layout of LHC machine and LEP are similar, having eight linear and eight arcs sections shown in Fig. 3.2 [36]. The LHC is designed to have two separate rings containing clockwise and counterclockwise proton beams, equipped with separate magnetic fields and vacuum chambers. These two rings only intersect each other at the points where detectors are installed. Point 1 and point 5, opposite to each other, where two general purpose detectors ATLAS and CMS are installed respectively. The other two important detectors are installed at point 2 (ALICE) and point 8 (LHCb) designed to study heavy ion and B Physics respectively. The two protons beams are injected in the vertical plane in both rings close to point 2 and point 8. Among 8 sections, there are four with no beam crossing at all. Point 3 and point 7 are equipped with two independent collimation systems for both rings. At point 4, two RF system are installed one for each beam. While at point 6, kicker magnets are installed to dump the beams.

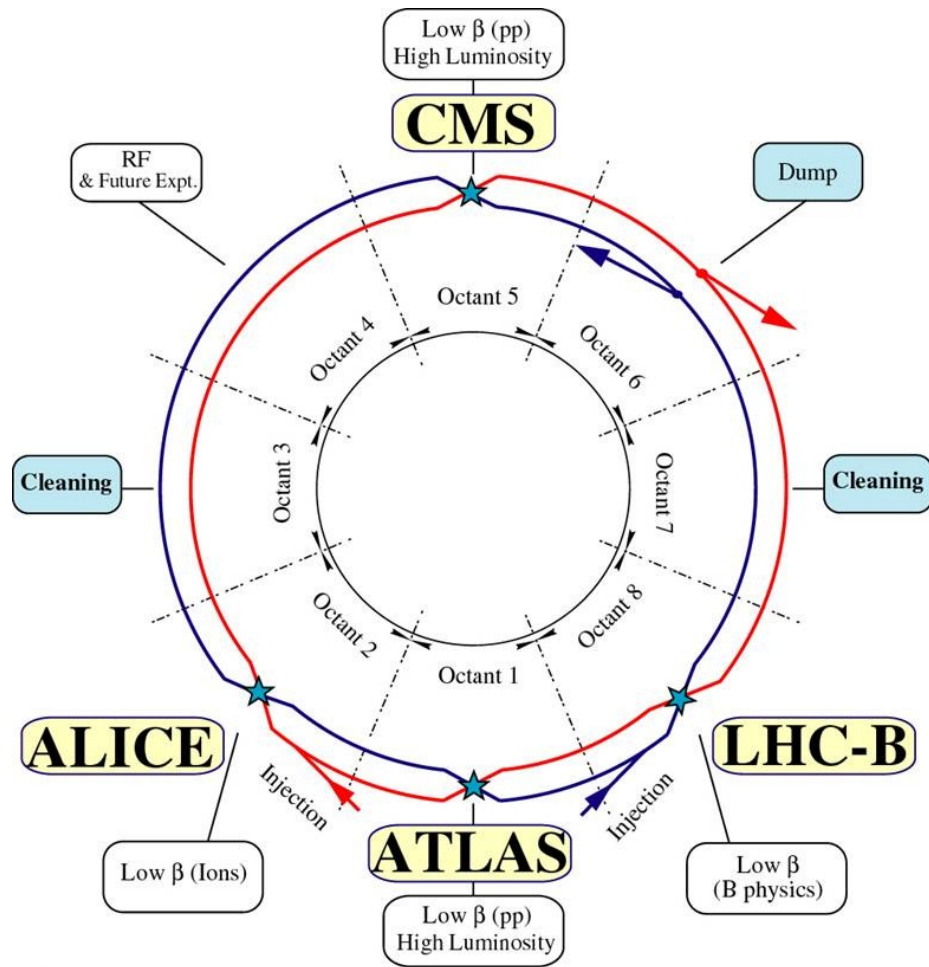


FIGURE 3.2: Schematic layout of the LHC

Performance Goals

The aim of the LHC to test Standard Model Physics to ultimate precision and hunt for any kind of new physics up to 14 TeV. The total total number of events (N) of a certain process produced at LHC can be given as

$$N = \mathcal{L} \sigma_{process} \quad (3.1)$$

Here $\mathcal{L}$, σ represents machine luminosity and the cross section for the process respectively. Cross sections for all processes produced by proton-(anti) proton collisions at machine luminosity $\mathcal{L} = 10^{33} \text{ cm}^2 \text{ s}^{-1}$ are shown in Fig. 3.3. The Higgs boson production at LHC is rare as compared to other QCD and weak boson (W, Z) productions which act as backgrounds in the Higgs boson searches. In order to reconstruct Higgs boson at LHC, one need to exploit specific final states providing enough signal to background discrimination. From Fig. 3.3 its clear that Higgs boson production rises with increase in center of mass energies. So increase in both machine luminosity and center of mass energies enables us to get more statistics of Higgs boson. The Nominal luminosity for CMS and ATLAS experiment is $\mathcal{L} = 10^{34} \text{ cm}^2 \text{ s}^{-1}$, and LHC nominal center of mass energy is $\sqrt{s} = 14 \text{ TeV}$. The cross section in Fig. 3.3 [37] shows that LHC has possibility to produce a single Higgs boson after every 100 s at mass 500 GeV. To get ultimate number of Higgs bosons produced at certain luminosity, one need to multiply this with specific branching ratio, selection and reconstruction efficiencies.

Nominal Luminosity and Beam Parameters

The machine luminosity is function of beam parameters, can be expressed as below

$$\mathcal{L} = \frac{\gamma_r N_b^2 f_{rev} n_b}{4\pi \epsilon_n \beta^*} F \quad (3.2)$$

here γ_r , N_b , n_b represents relativistic factor, number of particles per bunch and number of bunches per beam. While β^* represents the beta function at the collisions point and F is the factor to account for geometrical depletion in luminosity due to cross angle at interaction point (IP). This factor can be written as below

$$F = \left(1 + \left(\frac{\theta_c \theta_X}{2\sigma^*} \right)^2 \right) \quad (3.3)$$

where θ_c , θ_z and σ^* are crossing angle at interaction point, root mean square (RMS) bunch length and σ^* the transverse RMB beam size at at IP. The expression above is valid for two circular beams with same parameters.

TABLE 3.1: Parameters values to obtain LHC nominal luminosity for detectors

Parameter	Value
circumference	26.7 km
center-of-mass energy ($\sqrt{s}$)	14 TeV
Nominal luminosity	$10^{34} \text{cm}^2 \text{s}^{-1}$
Bunch spacing	25 ns
distance between bunches	7.5 m
Number of bunches in ring	2808
Number of protons per bunch	1.15×10^{11}
beta function at the collision point(β^*)	0.55 m
Transverse RMS beam size at impact point(σ^*)	$16.3 \mu\text{m}$
Dipole field at 7 TeV(B)	8.33 T
Dipole temperature (T)	1.9 K

The two major experiment at LHC, the CMS [38] and the ATLAS [39] are two general purpose detectors designed to operate at high luminosity of $\mathcal{L} = 10^{34} \text{cm}^2 \text{s}^{-1}$. The summary of the parameters to obtain this nominal luminosity is summarized in Table 3.1. Such high luminosity can not be attained by proton-anti-proton collisions, leaving only possibility of proton proton collisions for LHC machine. To align two opposite moving proton beams in both rings for collision needs dipole magnetic with opposite fields direction. The LHCb [40] detector is specially designed to study B Physics at peak luminosity of $\mathcal{L} = 10^{32} \text{cm}^2 \text{s}^{-1}$. At last, the ALICE [41] is the detector to study heavy ion collision targeting peak luminosity of $\mathcal{L} = 10^{27} \text{cm}^2 \text{s}^{-1}$.

3.1.2 Data Taking from 2010 to 2015

LHC started to deliver regular data from 2010. In 2010, LHC operated at 7 TeV center of mass energy with continuous increase in luminosity and delivered 47pb^{-1} proton proton collision data. During 2011, LHC continued with 7 TeV center of mass energy with higher luminosity as compared to last year and delivered 6.1fb^{-1} of data. Fig. 3.4a shows the LHC delivered and CMS recorded integrated luminosity throughout the year. LHC team decided to increase center of mass of energy to 8 TeV during 2012 and luminosity reached at the value of $10^{33} \text{cm}^2 \text{s}^{-1}$. LHC delivered 23.3fb^{-1} of data during 2012 as shown in Fig. 3.4b. The protons bunch spacing was reduced from 50 ns to 25 ns to increase the luminosity. The machine was shut-down for two years in 2013 and 2014 for major upgrades in LHC machine and

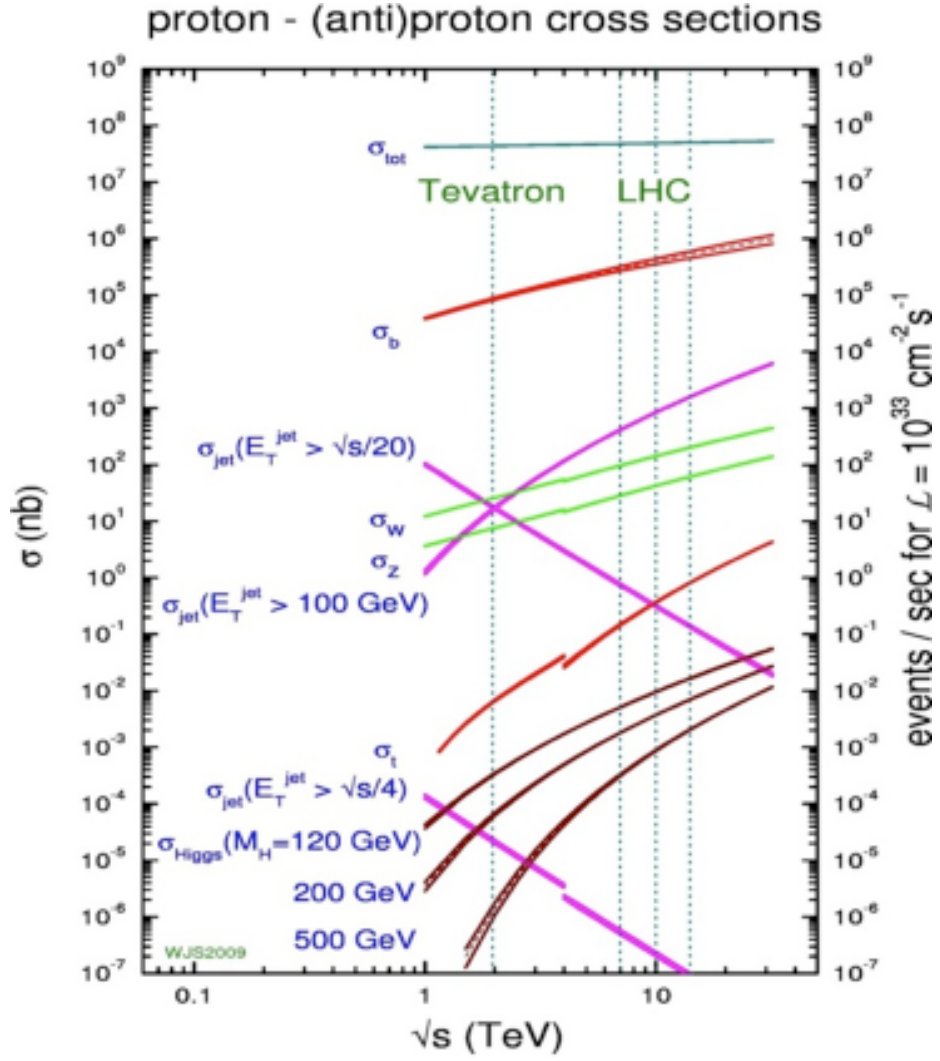


FIGURE 3.3: Cross section value for different process corresponding to machine luminosity $\mathcal{L} = 10^{33} \text{ cm}^2 \text{ s}^{-1}$.

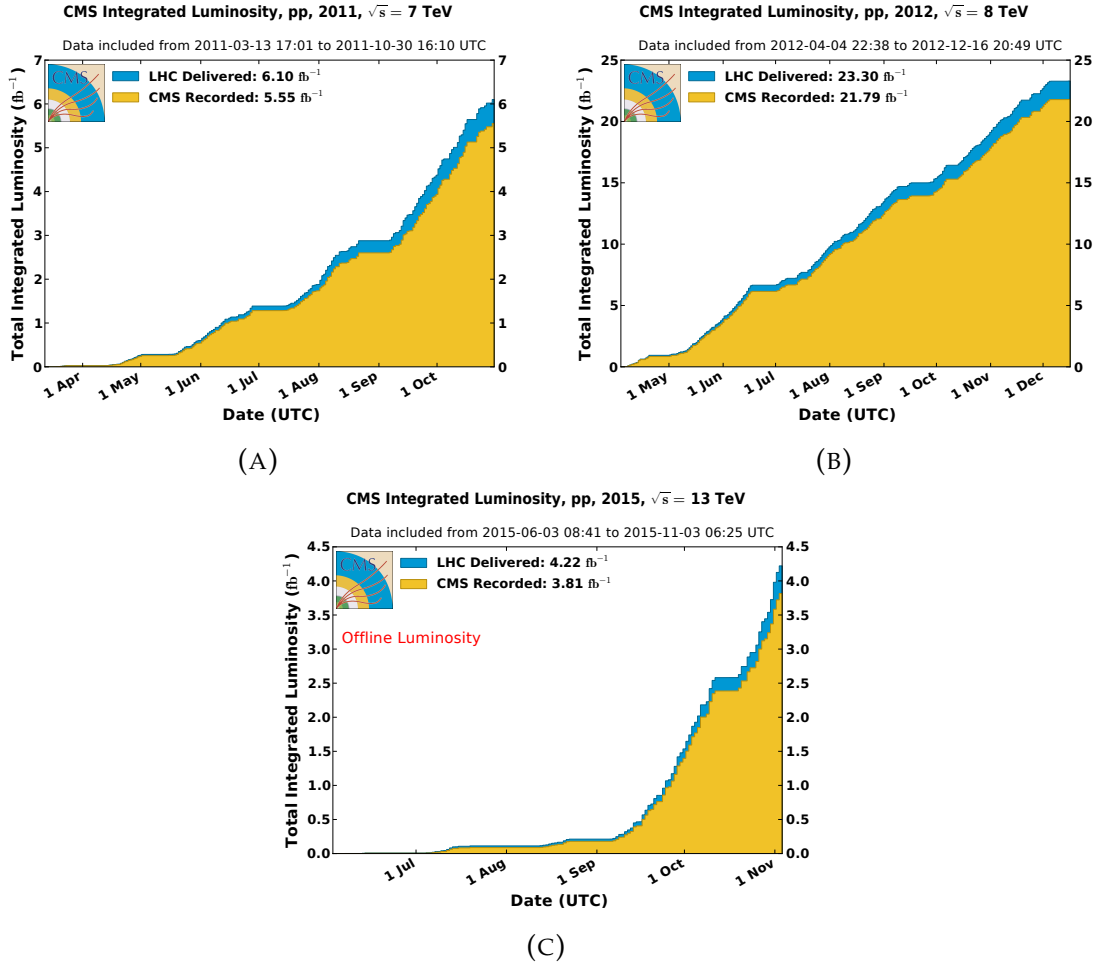


FIGURE 3.4: Integrated luminosities of LHC delivered (blue) and CMS recorded (yellow) data for 2011(a), 2012(b) and 2015(c)

detectors. LHC started again in 2015 at center of mass energies to 13 TeV (close to its design value i.e 14 TeV). LHC delivered 4.2 fb^{-1} –8.4 fb^{-1} during 2015 at 13 TeV center of mass energies shown in Fig. 3.4c.

3.2 The Compact Muon Solenoid Detector

The Compact Muon Solenoid is 28 m long, 15 m high and 14,000 tonnes heavy general purpose particle detector installed on LHC ring. CMS detector is capable to study a large range of physics from Standard Model Physics (including Higgs Physics) to search for extra dimension and the dark matter. CMS is efficient enough

to cover a large range of physics with energy ranges from a few 100 GeV to TeV scale [42]. CMS and ATLAS experiment shares the same physics goal though both independent experiment uses different technical solution and detectors design. Fig. 3.3 shows that the total proton-proton cross-section at center of mass energy $\sqrt{s} = 14$ TeV is approximately 100 mb. Keeping in mind the nominal luminosity will lead to have approximately 10^9 inelastic events/s. To deal with such high rate collisions is a experimental challenge. CMS trigger system helps to select online events and reducing the huge rate to approximately 100 events/s to be stored and used for physics analysis later. The read-out and trigger system design meets the short bunch crossing time, 25 ns.

Being CMS as general purpose detector, followings are main requirement to fulfill physics goal of LHC:

- Good identification and momentum resolution of muons covering wide range of momenta and angles and determination of the charge on muons with ultimate precision with momentum up to 1 TeV.
- Healthy inner tracker reconstruction efficiency and resolution of charged particle momentum. Efficient triggering and discrimination of τ 's and b-jets.
- Good dielectron and diphoton mass resolution and electromagnetic energy resolution.
- Good dijet-mass resolution and missing-transverse-energy (MET).

One of the main feature of a particle detector is the choice of its magnetic field configuration. Stronger magnetic field is needed to measure high energy charged particle momentum with precision. In CMS, this is achieved by selecting superconducting magnets.

CMS overall design is shown in Fig. 3.5. A 13-m long and 6-m inner-diameter superconducting solenoid producing a magnetic field of 3.8-T is placed at the heart of CMS. The inner tracker, electromagnetic and hadronic calorimeters both are fitted inside the large bore of magnet. Inner tracker of CMS have large track multiplicity as being closest to beam pipe where collisions happens. The electromagnetic calorimeter (ECAL) is made of lead tungstate crystals that covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Finally, Muon system is installed at most outer layer of CMS detector. Details of subdetectors, trigger and data acquisition systems would be described in next sections.

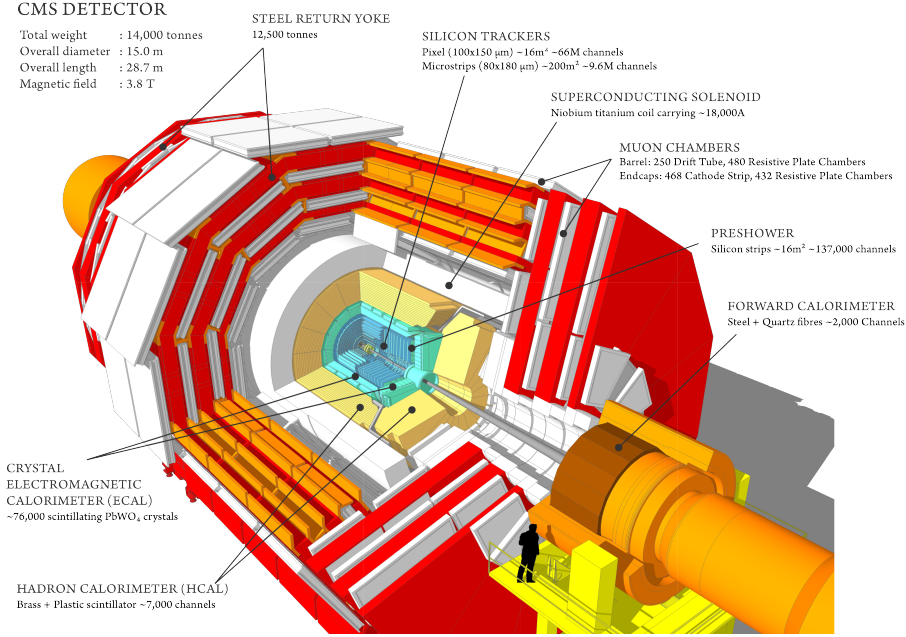


FIGURE 3.5: The CMS detector sectional view

3.2.1 CMS Detector Geometry

The coordinates system would be described in this section and will be used throughout. The CMS detector is like a closed cylinder around beam axis (Z axis) as illustrated in Fig. 3.5. The center of the experiment lies inside beam pipe where two beam collides; the y axis is directed vertically upwards; and the x axis points horizontally towards the center of the LHC ring. In the transverse (xy) plane, the radial component is represented by r and the azimuthal angle measured from the x axis is denoted by ϕ . The polar angle θ is measured from the z axis shown in Fig. 3.6. The direction of any particle in CMS detector is described by (η, ϕ) , where η is called pseudorapidity and defined as $\eta = -\ln \tan(\theta/2)$. As the LHC is protons collider, which contains partons (quarks and gluons) that carry different fractions of proton's total momentum. As a results, events in CMS detector will have boosted center of mass as compared to lab frame along z axis. New defined quantity pseudorapidity is invariant under these boosts. In short, pseudorapidity is approximation of rapidity in case of high energetic particles which are assumed to be massless and is defined below

$$y = \ln\left(\frac{E + p_z}{E - p_z}\right) \quad (3.4)$$

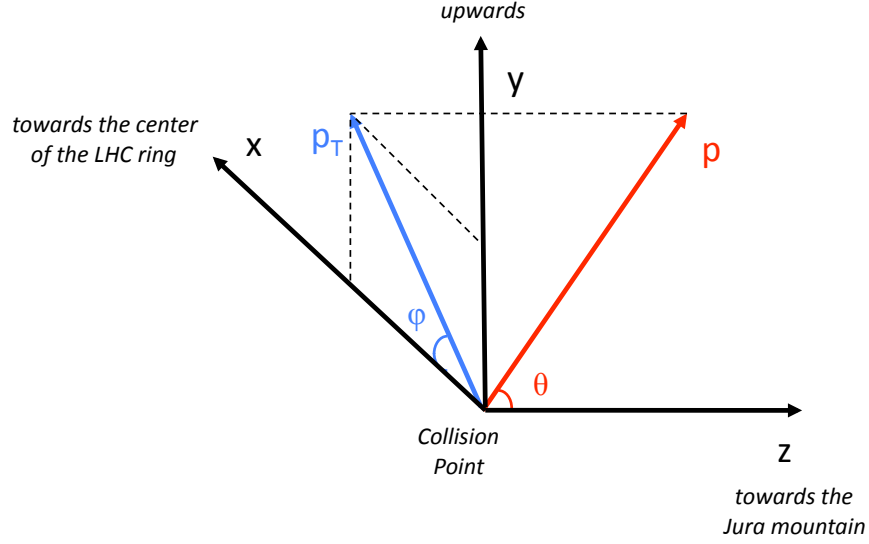


FIGURE 3.6: The CMS Experiment coordinate system

$$\approx \ln\left(\frac{1 + \cos \theta}{1 - \cos \theta}\right) \quad (3.5)$$

$$= \ln(\cot^2 \theta). \quad (3.6)$$

Pseudorapidity also helps to distinguish the barrel (central) and endcaps regions of subdetectors. The barrel and endcap regions are defined by $|\eta| < 1.5$ and $1.5 < |\eta| < 3.0$ respectively. Formally, electromagnetic calorimeter and tracking system provides a coverage a region up to $|\eta| < 2.5$, while muon station covers up to $|\eta| < 2.4$. So in CMS muons and electrons are selected with $|\eta| < 2.4$ and $|\eta| < 2.5$ respectively. As there is no transverse momentum (along y axis) before collisions leads to have transverse momentum approximately zero. The transverse energy of any particle is orthogonal to z axis, defined as $E_T = E \sin \theta$ and similarly the transverse momentum $p_T = p \sin \theta$. For a massless particle approximation, electrons and muons, transverse momentum is equal to transverse energy i.e $E_T = p_T$. As the CMS detector is hermetic, the missing transverse energy is understood as the energy of those particle which escaped from detector. These particles can be new physics particles, neutrinos and neutralinos which have no or very little interaction with detector material.

TABLE 3.2: Types of detectors installed in CMS tracking system

Section	Radial distance (cm)	Detector type	Cell size
Pixel	$r < 10$	pixel	$100\mu m \times 150\mu m$
TIB+TID	$20 < r < 55$	microstrips	$10cm \times 150\mu m$
TOB+TEC	$55 < r < 110$	strips	up to $25cm \times 180\mu m$

3.2.2 Inner Tracking System

The CMS inner tracking system is most inner sub-detector surrounding the beam pipe and interaction region as shown in Fig. 3.5. It has diameter of 2.5 m and a length of 5 m feeding $|\eta|$ coverage up to 2.5. The purpose of inner tracking system is to measure the charged particles trajectories needed to calculate their momentum and reconstruction of secondary vertices. The whole tracker is placed in a uniform magnetic field of 3.8-T provided by CMS solenoid. At LHC nominal luminosity of $10^{34} cm^2 s^{-1}$, there will be more than 20 proton-proton interactions leading to average about 1000 particles after every bunch crossing, 25 ns. To distinguish the trajectories and assign correct bunch crossing, fast response and high granularity detector technology is needed. To protect against radiation damage caused by intense particle flux and priority to minimize the material to avoid from multiple scattering, nuclear reaction, bremsstrahlung and photon conversion, a compromise was found to choose silicon detector technology. CMS inner tracker is largest silicon tracker ever made with about 200 m^2 of active silicon.

CMS tracker detector should be robust which can reconstruct tracks of charged particle in whole pseudorapidity range with momentum above 1 GeV to achieve LHC physics goals. To distinguish heavy flavors efficient reconstruction of secondary and impact parameter is necessary. Tracker also helps to identify other particles like electron, muons along with electromagnetic calorimeter. CMS inner operate under intense particle flux with hit density varies from 1 MHz/mm² at radial distance of 4cm decrease to 60 KHz/mm² at radial distance of 22cm and 3 KHz/mm² at distance of 115cm. Keeping hit densities in view, different types of detectors are installed at various radial distances of inner tracker listed in Table 3.2. Silicon detector are chosen due its unique properties of fast response and good spatial resolution which enables it to work efficiently even in high radiation environment. To limit the damage mainly by radiation the whole tracker system is kept at about -20° C.

The CMS inner tracker schematic drawing is shown in Fig. 3.7 [42]. Tracker most inner region is pixel detector that is nearest to the interaction point consists of two disks modules on each side and three barrel layers. It provides accurate traces in $r - \phi$ and z with a pixel cell size of $100 \times 150\mu m^2$ covers the $|\eta|$ range

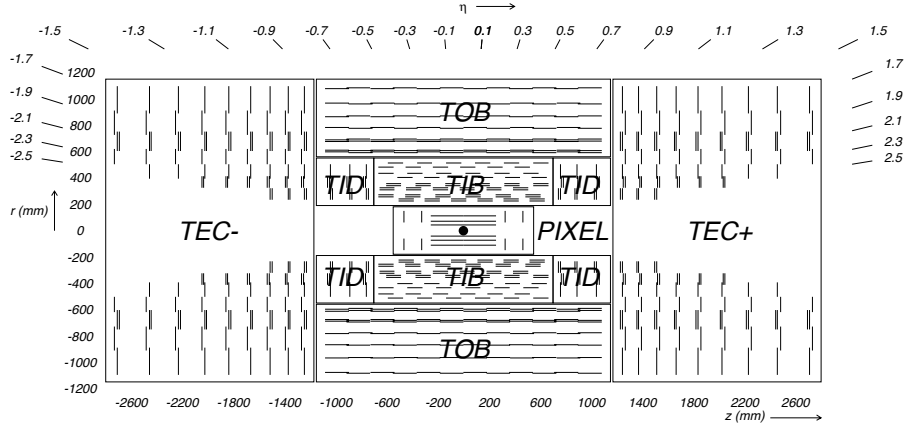


FIGURE 3.7: Schematic diagram of CMS tracker system. Lines represent detector modules.

up to 2.5. Pixel detector also plays its role for impact parameter resolution that helps to reconstruct secondary vertex. Pixel detector delivers three spatial point with excellent precision throughout the coverage of pixel illustrated in Fig. 3.8 [42]. Overall, the pixel detectors is of 1 m^2 size containing 66 million pixels. Second layer detector after pixel is silicon strip detector installed at radial distance from 20cm to 116cm. This detector surrounds both sides of pixel with its tracker inner barrel (TIB) and inner tracker disk (TID) regions. There are 3 disks on each endcap and 4 barrel layers where silicon microstrips of thickness $320 \mu\text{m}$ are installed. Microstrips are aligned along with beam axis in barrel region and in radial direction on the disks. The strip pitch in barrel region varies from $80 \mu\text{m}$ to $120 \mu\text{m}$ from inner to outer barrel layer. In the endcap region strip pitch changes from $100 \mu\text{m}$ to $141 \mu\text{m}$. Finally, the most outer part of CMS tracker system is tracker outer barrel (TOB) installed at radial distance from 55cm to 116cm containing $500 \mu\text{m}$ thick 6 barrel layers with strip pitch of $183 \mu\text{m}$ and $122 \mu\text{m}$. TOB provides z coverage up to 118 cm, beyond this point 9 disks of tracker endcaps (TEC+, TEC-) are located that covers $124\text{cm} < |z| < 282\text{cm}$ and $22.5\text{cm} < |r| < 113.5\text{cm}$.

To measure the momentum in CMS tracker, we simply exploit the property of bending of the charged particles in the magnetic field. High momentum particles will bend less in detector and vice versa, making resolution worse due to less curvature. The material in radiation length's for different subdetectors and functional modules are shown in Fig. 3.9 [42]. As in tracker, strips and pixel sizes are not uniform with the variation of η , Fig. 3.10 [42] shows the expected performance of muon tracks resolution with $p_T = 1, 10$ and 100 GeV as function of $|\eta|$. While Fig. 3.11 [42] shows comparison of global track reconstruction efficiencies of

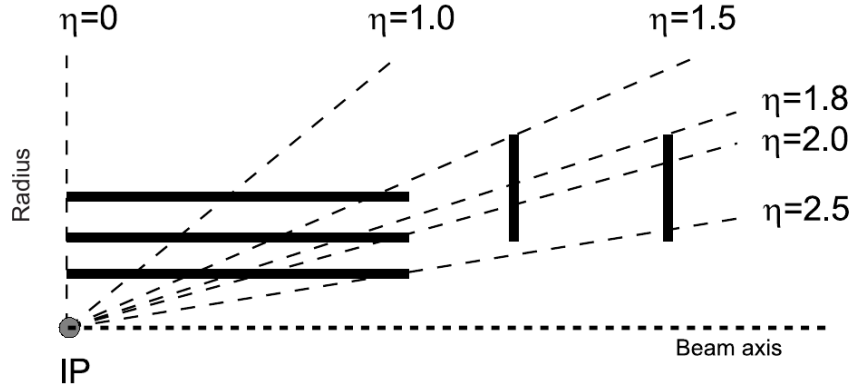


FIGURE 3.8: CMS pixel detectors and hit coverage. Lines represent detector modules.

muons and pions with same transverse momentum.

3.2.3 Electromagnetic Calorimeter

Electromagnetic calorimeter (ECAL) is the cylindrical shape hermetic homogeneous calorimeter made of lead tungstate (PbWO_4) crystals. Barrel region of the ECAL (EB) is providing pseudorapidity coverage of $|\eta| < 1.479$ and contains 61200 PbWO_4 crystals. Endcap region of the ECAL (EE) is providing pseudorapidity coverage up to $\eta < 3$ containing 7324 crystals in each of the two endcaps. The crack region between barrel and endcaps ($1.4442 < |\eta| < 1.56$) as shown in Fig. 3.12 [42] is excluded for physics analysis. A preshower detector is installed in endcap regions to distinguish between single and diphoton close to each other e.g neutral pion decay ($\pi^0 \rightarrow \gamma\gamma$). Specially designed Avalanche photodiodes (APDs) and Vacuum phototriodes (VPTs) are used in barrel and endcaps region respectively ensure efficient working under high magnetic field. One of the ECAL key feature in the design is to reconstruct electrons and photons with high resolution enabled us to discover Higgs boson in both $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ \rightarrow 4\ell$ favorable channel.

ECAL Crystals and Geometry

To make a compact and high granularity calorimeter which could fit inside the CMS solenoid, lead tungstate was the best option which is denser than steel (8.28 g/cm^3) with short radiation length (X_0) of 0.89 cm [43]. The scintillator (PbWO_4) emit light directly proportional to the energy of the incident particle through ionization. Fast response (same order of LHC bunch cross crossing i.e. 25ns) of

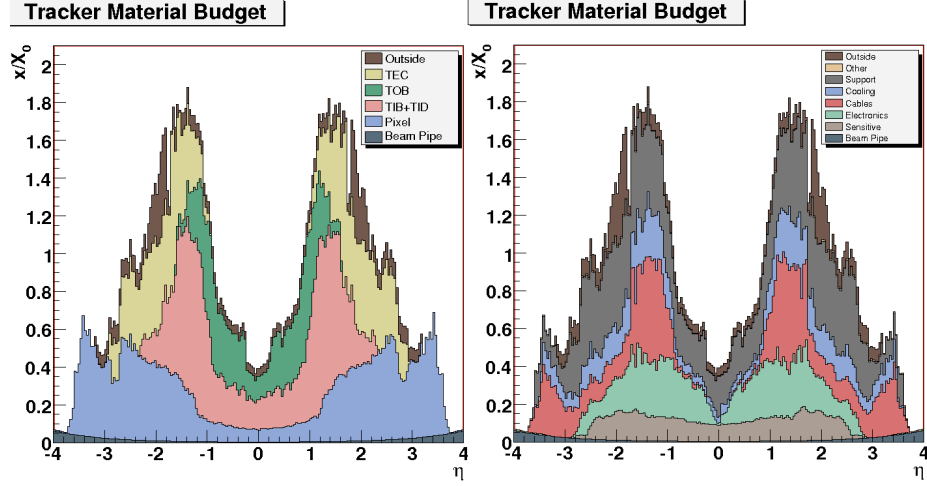


FIGURE 3.9: Material in radiation length's for different subdetectors (left) and for different functional modules (right) as function of η

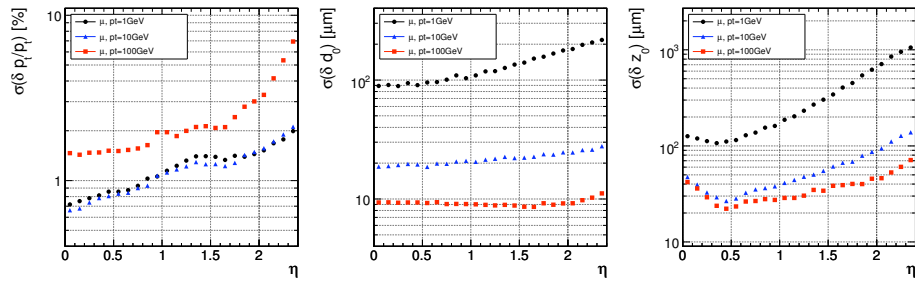


FIGURE 3.10: Expected single muons tracks resolution of transverse momentum (left), transverse impact parameter (middle) and longitudinal impact parameter (right) of $p_T = 1, 10, 100$ GeV.

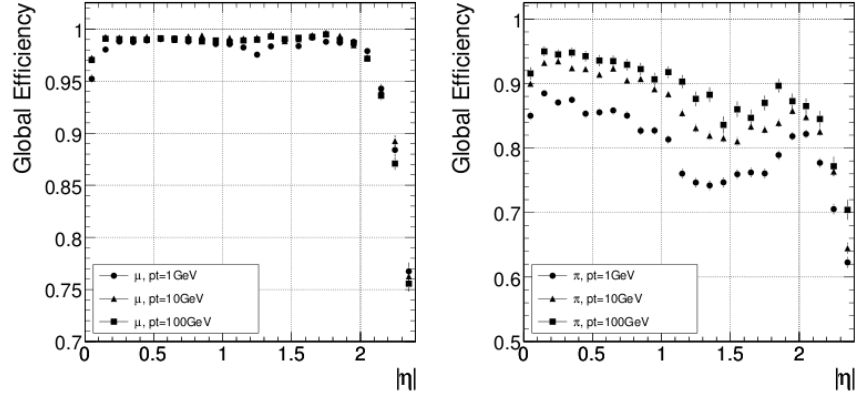


FIGURE 3.11: CMS tracker global efficiency of muons (left) and pions (right) of $p_T = 1, 10, 100$ GeV.

TABLE 3.3: ECAL crystal details at endcaps and barrel region.

	Barrel	Endcaps
Number of crystal	61200	14648
Crystal cross section in (η, ϕ)	0.0174×0.0174	variable
Cross section of front face of crystal	$22 \times 22 \text{ mm}^2$	$28.62 \times 28.62 \text{ mm}^2$
Cross section of rear face of crystal	$26 \times 26 \text{ mm}^2$	$30 \times 30 \text{ mm}^2$
Crystal length	230 mm	220 mm
Crystal length(in terms of X_0)	$25.8 X_0$	$24.7 X_0$

PbWO_4 makes it ideal for ECAL. In barrel region, there are 8.14 m^3 active crystals installed weighting up to 67.4 tonne. The each crystals cross section is measured to be $22 \times 22 \text{ mm}^2$ in barrel and $28.62 \times 28.62 \text{ mm}^2$ at endcaps. The length of the crystals in barrel and endcaps is 230mm and 220 mm which is 25.8 and 24.7 times of the radiation length.

The amount of emitted light from scintillator amplification of photodiodes both are temperature dependent, the nominal temperature of ECAL operation is 18°C . For this reason, cooling system has been installed in ECAL to maintain the temperature at nominal value. The cooling system rotates water at 18°C through parallel grid of aluminium pipes a to steady the temperature in detector. In the barrel region, each supermodule has its independent supply of water. The cooling system of the ECAL work efficiently overall and maintain the temperature of $18 \pm 0.05^\circ\text{C}$, hence keeping energy resolution steady.

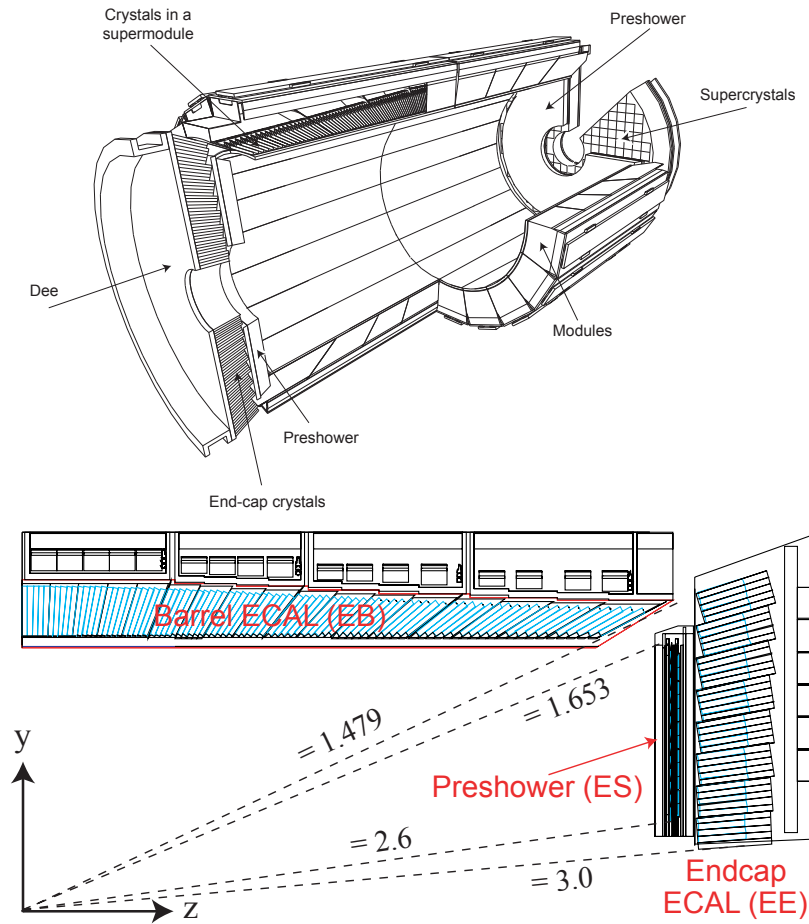


FIGURE 3.12: CMS ECAL layout and pseudorapidity coverage.

Photodetectors

The requirement on photodetectors are to be fast, robust and able to work efficiently in a region that contains 3.8-T magnetic field. Considering these requirement, different types of photodetectors are installed in the barrel and the endcaps regions. In the barrel region, a pair of avalanche photodiodes are mounted with active area of $5 \times 5 \text{ mm}^2$ on each crystal. APD are working at gain 50 and in parallel read out. In the endcap region, a specially designed vacuum phototriodes at National Research Institute Electron in St. Petersburg for CMS ECAL was used. These are single gain photomultipliers with mean gain of 10.2 at zero field. The diameter of each VPT is 22mm. The total active area is almost 280 mm^2 where one VPT is attached at the back of every crystal. Their fine copper anode makes them capable to work under intense magnetic field. As compared to APDs, VPTs has lower quantum efficiency and internal gain which is balanced out with their larger back face crystal coverage. The dimensions and number of crystals has been in the barrel and endcaps is given in Table ??.

Preshower

The purpose of the installation of preshower detector is to identify neutral pions in the endcap region. The preshower provides the pseudorapidity coverage of $1.653 < |\eta| < 2.6$. It also helps to distinguish between electron and minimum ionizing particles (MIPs) and the position determination of electrons and photon due to its high granularity.

Energy Resolution

The energy resolution of can be defined as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{S}{\sqrt{E}}\right)^2 + \left(\frac{N}{E}\right)^2 + C^2 \quad (3.7)$$

where C is constant term extracted by fitting of measured point to the function, N is noise term and S is stochastic term. A measurement of energy resolution has been made by using electron of energies from 20 to 250 GeV in test beam [44]. The measurement was made by considering arrays of 3×3 or 5×5 . No correlated significant noise was observed in these arrays. The measurement summary in shown in Fig.3.13 [42].

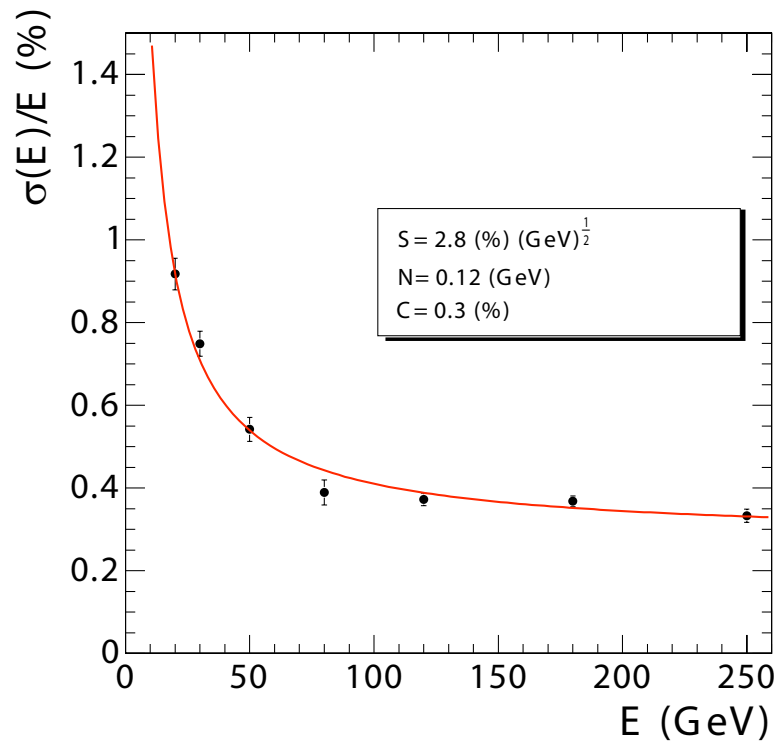


FIGURE 3.13: CMS ECAL expected energy resolution.

3.2.4 Hadron Calorimeter

CMS being a general purpose detector is capable to study several final state signatures involved in a large range of high energy processes. The hadron calorimeter (HCAL) provides the measurement of energies and positions of hadronic jets (for example neutrons, protons, kaons and pions) [45]. HCAL being hermetic is also helpful to calculate MET that can be assigned to neutrinos or other invisible particles. There are no gaps or cracks in HCAL to make sure to capture all known particles, leaving MET with only possibility of escaped particles. HCAL of CMS is fitted between ECAL and CMS magnet solenoid at a radial distance from 1.77 m to 2.95 m. HCAL provided a pseudorapidity coverage up to $|\eta| < 3.0$, beyond this point a forward hadron calorimeter (HF) is installed at a distance of 11.2 m from the collision point on both ends extending pseudorapidity coverage to $|\eta| < 5.2$. As HCAL is placed inside 3.8-T of magnetic field, so absorber must be a non-magnetic material. Also, due to requirement on maximum number of interaction length, cost and mechanical property brass has been chosen for CMS HCAL.

As usual HCAL can be splitted into barrel (HB) and endcap (HE) regions with barrel and endcaps pseudorapidity coverage of $|\eta| < 1.33$ and $1.3 < |\eta| < 3$ respectively. HB consists of 36 similar wedges each weighing 26 tonnes that makeup two half-barrels (HB+ and HB-) illustrated in Fig. 3.14. The wedges are made of plane brass absorber plates kept along beam axis. All plates are made of brass with physical properties described in Table ?? except inner and outer most made of steel for structural strength. Table ?? shows the thickness of each layer. HCAL uses the alternate layers of absorber brass and plastic scintillator to measure the particle position, energy and arrival time. The thickness of the brass plate are 79 mm with 99 mm gaps for scintillators installation. Optical fiber collect all the light emitted when a particle passes through scintillator and passed it to readout box. Total energy is calculated by summation of the light over many layers in depths called tower. The scintillator in HCAL are arranged in the form of 70,000 tiny tiles. individual tiles makes scintillator tray by grouping them in ϕ layers to limit the number of readout boxes. These trays works as individual units, so it can be replaced or tested easily. The focus of the design of HCAL endcap was to minimize the cracks between HB and HE rather than single-particle energy resolution.

Forward calorimeter

The two HF are placed on both sides of CMS to catch up the particle emerging from proton- proton collisions at very small angle relative to beam axis as shown in Fig. 3.14 [42]. HF is most forward region of the detector where enormous particle fluxes are expected. On average, HF absorbed 8 times more energy is deposited in two forward calorimeter as compared to whole CMS detector. The design of HF

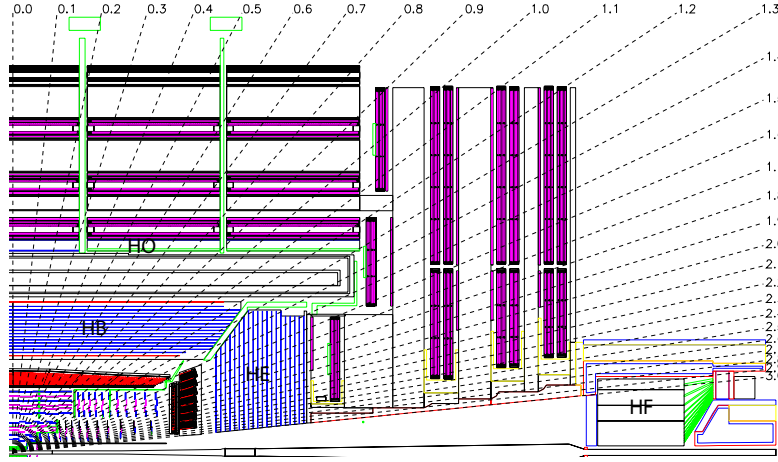


FIGURE 3.14: CMS HCAL layout.

TABLE 3.4: Properties of brass of HCAL.

chemical composition	70% Cu, 30% Zn
density	8.53 g/cm ³
radiation length	1.49 cm
interaction length	16.42 cm

calorimeter for such harsh condition was considerable challenge. HF will absorb a radiation dose of about 10 MGy after an integrated luminosity of $5 \times 10^5 \text{ pb}^{-1}$ at $|\eta| = 5$. The rate of charged hadrons is also extremely high. Because of hard condition HF material and design is different from rest of HCAL.

The energy resolution of the HCAL is given as

$$\left(\frac{\sigma(E)}{E}\right)^2 = \left(\frac{90\%}{\sqrt{E}}\right)^2 + (4.5\%)^2 \quad (3.8)$$

and for HF is given as

$$\left(\frac{\sigma(E)}{E}\right)^2 = \left(\frac{170\%}{\sqrt{E}}\right)^2 + (9.0\%)^2. \quad (3.9)$$

TABLE 3.5: HCAL absorber thickness

layer	material	thickness
front plate	steel	40mm
1-8	brass	50.5mm
9-14	brass	56.5mm
back plate	steel	75mm

3.2.5 Superconducting Magnet

The CMS central is superconducting solenoid [46, 47, 48, 49] designed in 6 m diameter and 12.5 m long bore producing magnetic field of 3.8-T which is 10,000 time stronger than earth magnet . The CMS solenoid is the largest ever built magnet weighs 12,000 tonnes, powerful enough to provide a strong magnetic field throughout the detector. The CMS inner tracker, ECAL and HCAL are fitted inside the bore of the magnetic field. The task of the magnetic field is to bend the charged particles when they pass through. More the momentum of the charged particles less they bend, so to measure the momentum of high energetic charged particles a stronger magnetic field is required. CMS aim to have excellent momentum resolution with such strong magnetic field specially for muons. The CMS magnetic field is a solenoid made of coil of NbTi wire which supplies the uniform magnetic field when current pass through it. The NbTi becomes superconductor at -268.5°C (4 K) which allows a huge amount of current (10,000 A) to pass through it. To maintain this temperature liquid helium is used. A large bending is achieved in tacker by stronger magnetic field providing precise and efficiencies momentum measurement. The magnetic field direction inside (tracker) and outside the solenoid (muon system) allows to make second momentum measurement for muons.

3.2.6 Muon System

Muon identification in CMS is provided by its outer most layer of sub-detector called muon system [50]. The detection of the muons is very important for many physics process. In particular, for Higgs boson SM decay to ZZ/ZZ^* which is labeled a golden channel if each Z boson decay to a pair of opposite charged muons. The best mass resolution can be obtained with four muons in final state as muons are less effected as compared with electrons by detector material because of it interacts only weakly. Along with this example, SUSY and other model for new physics

searches also encourages the necessity for reconstruction of muons with excellent resolution.

To have a robust and precise muons system was one of the core motive of CMS as indicated by detector name. Identification, triggering and momentum measurement of the muons are the tasks assigned to the muon system. The flux-return yoke and strong magnetic field are playing crucial role for excellent muons momentum resolution and triggering. The flux-return yoke by absorbing the hadrons helps for muons identification.

Muon system was essentially designed as cylindrical shape to provide full pseudorapidity coverage with detection area of 25000 m^2 , having barrel and endcaps regions. Muons detection is based on gasses detector of three types. In the barrel region, the magnet field is uniform with less neutron-induced background drift tube are installed providing pseudorapidity coverage up to $|\eta| < 1.2$, as shown in Fig. 3.15. The barrel region is arranged to have four muons station, measuring $r - \phi$ bending plane and distance along beam side. The arrangement of the drift cells are offset by the half of the width of the cell providing dead spot in efficiency, determination of standalone bunch crossing and excellent time resolution by utilizing meantimer circuits. The direction and number of chambers are optimized to get maximum efficiency of combination of the hits from different stations to get single track which is helpful to reject the background. In the endcap region, where non-uniform magnetic field and more background is expected cathode strip chambers (CSCs) are installed that providing coverage $0.9 < |\eta| < 2.4$. Like DT, there 4 CSCs stations on each endcap. Each station is composed of chambers, which are capable to work independently, aligned perpendicularly to z-axis and interspersed between the flux-return plates. To overcome the uncertainties in background rates and assignment of correct bunch crossing number, resistive plate chambers (RPCs) are installed both in barrel and endcap regions. There is a dedicated trigger system for each type of muons gasses detector including RPCs. Good time resolution, fast response and coarser spatial resolution makes RPCs distinct from CSCs and DT. The RPCs also plays vital role for the combination of tracks from multiple hits a chamber. The RPCs are installed in between of DT and CSCs region the barrel and endcap region.

Drift Tube chamber

The DT chambers are installed in the barrel region of muon system that provides pseudorapidity coverage up to $|\eta| < 1.2$ illustrated in Fig. 3.15. Whole DT system consists of 5 wheels along z-axis, and 4 stations along radial direction (MB1, MB2, MB3, MB4) each made of 12 chambers except fourth which contains 14 chambers. Each drift chamber of an average size of $2\text{m} \times 2.5\text{m}$ is made of 12 aluminium layers,

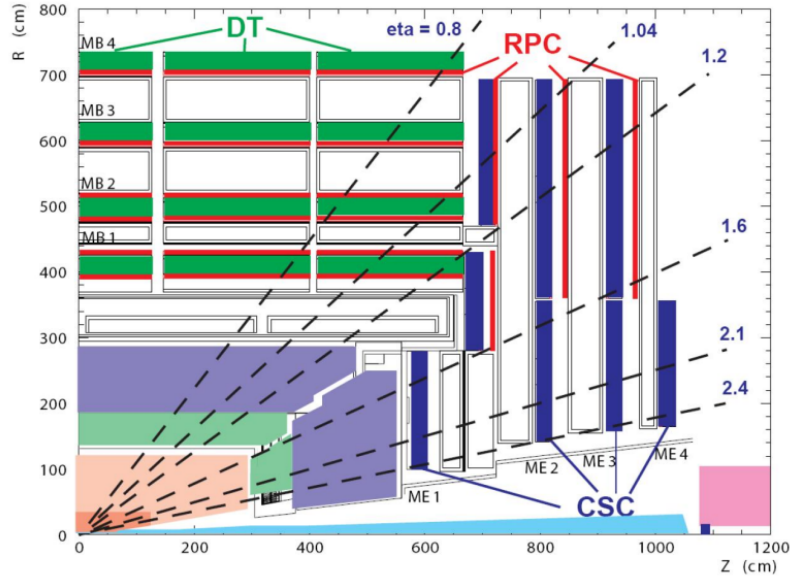


FIGURE 3.15: CMS muon system layout.

making 3 groups of four. The central group provides the coordinates measurement along z-axis and other two groups measures $r - \phi$. There are about 60 drift tubes in each chamber, each 4 cm wide made of drift cells each of size $42 \text{ mm} \times 13 \text{ mm}$. The $50 \mu\text{m}$ wire made of stainless steel serves as anode while two “I-shaped” aluminum tape each of thickness $50 \mu\text{m}$ are cathodes of the drift cell as shown in Fig. 3.16. As any charges particle pass by the drift cell, it ionizes the gas by removing electrons from its atom. This establish a electric field finishing of at positively charged wire. Distance between the wire and the muon track is estimated by drift time of these electrons. The cell is filled with gas mixture of CO_2 (15%) Ar (85%) which is inexpensive and safe. Other qualities of this mixture are good saturation drift velocity ($5.4 \text{ cm}/\mu\text{s}$) and quenching. To keep drift velocity steady, the filled gas is kept at a constant pressure inside the chambers. The pressure in each wheel is continuously monitored by sensors. It is expected to inject 10% new gas everyday.

Cathode Strip Chambers

The cathode strip chambers are installed at endcap region of CMS muon system providing pseudorapidity coverage of $|0.9| < |\eta| < 2.4$ illustrated in Fig. 3.15. The magnetic field in the endcap is non-uniform which forced to use different technology for barrel and endcap regions of muon system. Another challenge for CSCs is higher expected neutron-induced background as compared to the barrel region.

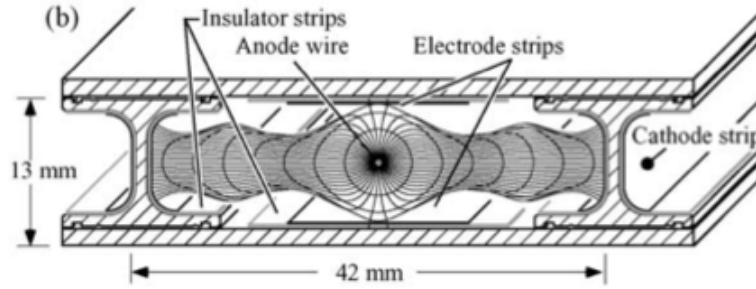


FIGURE 3.16: DT electric field created by cathode, anode and insulating strips .

The muon endcap is made up of 468 CSCs arranged in four station like the barrel region (ME1, ME2, ME3 and ME4). The pseudorapidity range from $|0.9| < |\eta| < 1.2$ is covered by both CSCs and DTs. The chambers are aligned trapezoidal, each chamber is covering 10° or 20° in ϕ . RPCs are also installed between CSCs stations.

The basic unit of muons end caps is a CSC which capable to work independently shown in Fig. 3.17a. The CSC is multiplewire detector made of 6 layers of positively charged anode interleaved with 7 layers of negatively charged cathode wires made of cooper filled within gas volume as shown in Fig.. The cathode and anode wires are perpendicular to each other and provides two position coordinates of each particle. As the charged particle pass through chamber, it will ionize the gas atoms producing avalanche of electrons to anode wires illustrated in Fig. 3.17b. As the wires are nearby, CSCs not only provides excellent momentum resolution but helps to trigger the event as well. It feeds $r - \phi$ resolution of 2 mm to first level trigger, $75\mu\text{m}$ off-line spatial resolution in $r - \phi$ for ME1/1 and ME1/2 chambers and $150\mu\text{m}$ for all other stations. The CSCs also help to assign bunch crossing with an efficiencies of 92% which can be enhanced up to 99% by simple majority out of all chambers. The tendency to operate at high rates and non uniform magnetic field makes CSCs robust which does not require precise gas pressure and temperature. Multilayer CSCs is useful for the combination of hits in CSCs and tracker detectors.

Resistive Plate Chambers

Resistive plate chambers are installed in CMS muon system in both barrel and endcap region. The motivation of RPCs are to get excellent time resolution along with good spatial resolution. The RPCs are very fast responsive detector providing

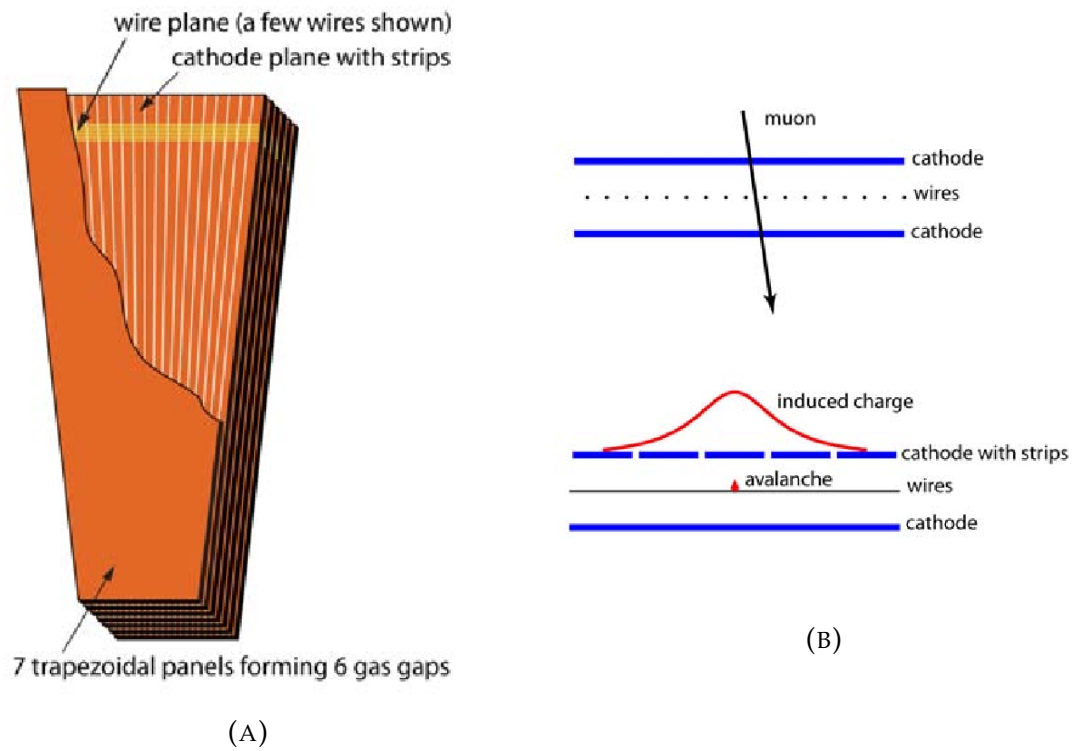


FIGURE 3.17: Multilayer CSCs layout and induced current in it.

another muon trigger along with CSCs and DTs. RPCs are capable to provide a time resolution of just one nanosecond.

RPCs are made of two high resistive plastic positive and negative charged plates, kept parallel to each other, with gas filled in between them. The gas mixture consists of 95% of $C_2H_2F_4$ and 5% of C_4H_{10} . As the muon pass through the RPCs, primary and secondary ionization of the gas create avalanche of electrons which is picked up by external metallic strips after a small but precise time delay. Quick momentum measurement is feeded to trigger the event. The RPCs excellent time resolution assign bunch crossing with ultimate precision.

Momentum Resolution

CMS detector is capable to measure muons transverse momentum independently by using tracker or muon system. The expected transverse momentum resolution of muons system has been studied for the using muons of $p_T = 40$ GeV from Z and W boson decays and found to be 10%(20%) in barrel and endcap regions. Global muons can also be reconstructed at CMS by the combination of both tracker and muons system. For muons with p_T below 100 GeV tracker provide better resolution than muon system and vice versa. The transverse momentum resolution found to be improved sufficiently by using global muons i.e 1%(2%) for $p_T < 100$ GeV in barrel(endcap) regions. For muon with $p_T > 100$ GeV, the transverse momentum of muons system improves to 8% (10%) in the barrel and endcap region. The momentum resolution of CMS muon system alone, tracker alone and combination of both are shown in Fig. 3.18.

3.2.7 Trigger and Data Acquisition System

The LHC running at it nominal parameters provides collisions at the rate of 40 MHz. Most of these events are soft collisions and uninteresting for new physics as they are like minimum bias collisions. Though CMS detector is capable to reconstruct all of these events but store all these events will fill the CMS storage in days. So trigger system of CMS helps of select interesting events out of all for further steps. The events which does not pass the first step trigger are deleted permanently. Trigger system performs the first step for physics event selection which reduces the rate from 40 MHz to 300KHz into two layers Level-1 (L1) trigger and high-level trigger (HLT). Before trigger selections, all events are kept so the trigger must very fast and robust.

The L1 trigger is made of programmable electronics, dropping the rate drastically to 30 KHz. To be fast, L1 trigger events by their energy and momentum calculations estimated by only using coarse in muon system and calorimeter instead

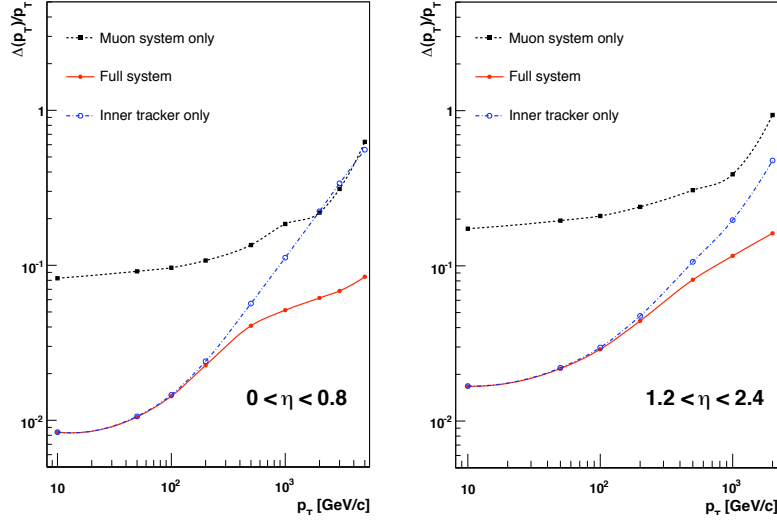


FIGURE 3.18: Muon momentum resolution as a function of p_T of muons in barrel and endcap regions.

of complete high resolution information. The L1 trigger decides to select or reject event in $3.2\mu\text{m}$.

All events with positive L1 decision with their complete information are transferred to HLT which is a farm of one thousand commercial processors. It uses the algorithms with growing complexity that is based on fine granularity of the event. HLT decision time is different for each event with mean time of about 50 ms. The HLT has excess of complete information of the event that demands a greater bandwidth of the order of 1Tb/s.

For example, to look for Higgs boson decay into 4-leptons final state, the trigger would look for energetic leptons in the detector. For L1 trigger, electron and muons points to a narrow energy deposit in ECAL and hit pattern in tracker and muon system respectively. While for HLT, the total energy and momentum of the leptons are reconstructed by ECAL and combination of muon system and inner tracker.

Level-1 Trigger Architecture

The configuration of L1 trigger is shown in Fig. 3.19. L1 trigger can be divided into calorimeter and muon triggers. Each ECAL and HCAL has independent regional trigger feeding to global calorimeter trigger. Similarly, muon trigger has local triggers for each subdetector merging into global muon trigger. Finally, global

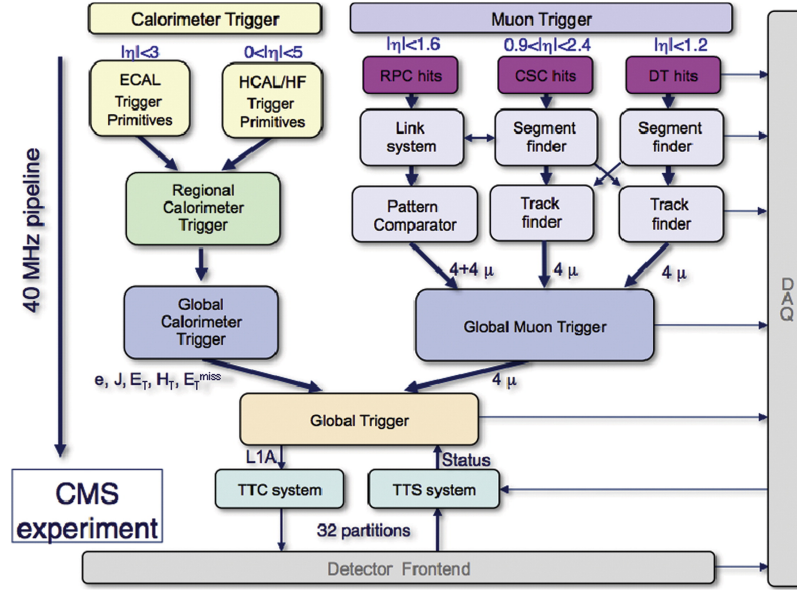


FIGURE 3.19: The configuration of L1 trigger.

calorimeter and muon triggers merge into global trigger which makes L1 final decision.

Each local trigger feeds to global/regional trigger by providing trigger primitives groups (TPGs) based on coarse of sub-detector. Regional calorimeter trigger (RCT) builds objects which deposit their energy in ECAL and HCAL such as jets, γ , electrons, photons and tau's. Similarly, in muon trigger, DT/CSC/RPC track finders feeds muons tracks to muon global trigger (MGT). At most four best candidates are sent to global muon and calorimeter trigger which further sent to global trigger (GT). GT compares these candidates with L1 trigger menu to decide weather to keep to reject the event. Some triggers in the menu evolved with increasing center of mass energy of LHC by tightening the requirement on candidate or be pre-scaled (selecting only the n th event) to control the output rate. Candidates able to fire atleast one trigger out of whole trigger menu is kept for further investigation by HLT. For example, the trigger/bit 'L1 SingleEG8Iso' in the L1 trigger menu requires at least one isolated (candidate with no little hadronic activity around it) electron or photon with transverse momentum greater than 8 GeV.

High-Level Trigger Architecture

HLT is three level (2, 2.5 and 3) trigger with growing complexity starting at level 2 which usually works on top of the information provided by L1 trigger. HLT

level 2 is to build fine granularity object by combining the information from sub-detector e.g from preshower and ECAL energy clusters are combined for electrons. Bremsstrahlung of the electrons causes its energy to spread into several neighbor cells. To compensate this effect, clusters of ECAL are combined to superclusters.

At HLT level 2.5, the inner tracker information is combined and matched with calorimeter and muon system. In case of electron, energy supercluster is matched with the hits in the tracker, if matches it form a seed. For muon, hits in the tracker are matched with the hits in muon system to reconstruct global muon. HLT level 3 reconstructs the complete track, if its found electron seed or matching hits of muon. After these three stages, precise energy and momentum are calculated of the objects that decides to accept or reject the event.

There are more than 200 different HLT paths each derived by at least one L1 bit. The data which pass the HLT trigger is organized into different datasets according to the objects reconstructed in the event and the passed selection e.g an event with two muons reconstructed with transverse momentum greater than 8 and 17 GeV will be placed in "DoubleMu" dataset.

Data Acquisition System

The schematic skeleton of CMS Data Acquisition (DAQ) system is shown in Fig. 3.20. CMS DAQ duty is to provide data from 650 different readout channels to HLT along with enough computing resources to limit the output data rate [51]. An event fragments with a size of 2KB is provided by each module. The CMS DAQ structure is outlined in Fig. 3.20. A network with a bandwidth of 100Gb/s is utilized by Front end drivers (FEDs) to read the data from detector sensors. These FEDs are located underground at the level of the detector called counting room. Events are transferred to the event filter systems at a maximum rate of 100 kHz. The complete reconstruction of objects of events and high machine luminosity is the reason for this high rate.

3.2.8 Computing Grid

One important and last step after data collection is to process it with full detailed reconstruction algorithms and made available for physics analysis. Instead of trigger system of CMS drastically decrease the amount of data but even then CMS delivers data at the order of petabytes every year. To analysis and store this huge amount of data, LHC has connected huge number of computing resources from collaborative institutes/universities, computer centers and middleware providers through the Worldwide LHC Computing Grid (WLCG).

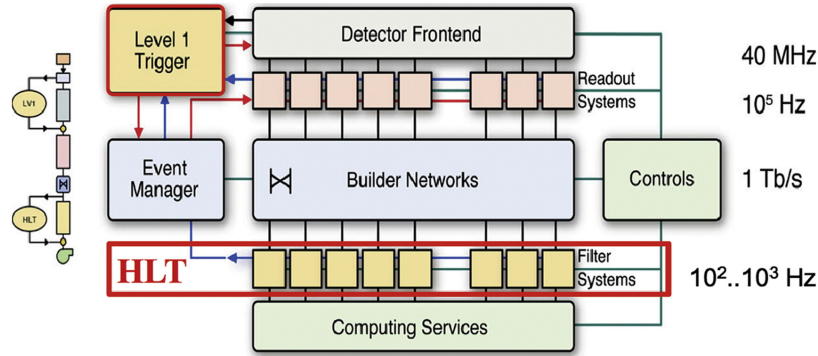


FIGURE 3.20: The CMS DAQ system structure.

There are three layers of tier (tier 0, tier 1 and tier 2) for data flow between different computing sites. The types of dataset is described below briefly

- RAW that contains all the recorded information from detector along with triggers decisions and other metadata. A copy of RAW data is achieved and placed at permanent storage element. A occupancy of around 1.5 (2.0) MB/event is needed to store data (simulated data) in RAW version.
- Many algorithms and pattern recognition like cluste and track-finding; primary and secondary vertex reconstruction etc are applied to RAW to obtain RECO data format. RECO events occupancy is roughly three times less than RAW (0.5 MB/event).
- The most compact format of data AOD (analysis object data) is obtained by dropping unnecessary information and reducing per event size up to 100KB.

Tier 0 is located at CERN with primary duties to copy recorded data to permanent storage, promptly reconstruct data from RAW version to have first version of prompt-RECO data and exporting a copy of RAW data to tier 1 centers. Tier 1 are mainly hosted by national computing centers and labs from collaborative countries. At each Tier 1, a fraction of detailed the reconstruction of RAW data simulated data has been performed to get AOD. Tier 1 also responsible to store and export AOD format data to tier 2. The last layer tier 3 is hosted by CMS collaborative institutes where main analysis activity, alignment, detector related studies and offline calibration is done. Tier 3 are used for Monte Carlo (MC) data production which can be exported to tier 1 for long term storage.

Chapter 4

Physics Objects

4.1 Muons

4.1.1 Muon Reconstruction

Standalone muon reconstruction

Global muon reconstruction

Cosmic-ray and beam-halo muon reconstruction

Performance

4.1.2 Muon Identification

Algorithm

performance

4.1.3 Muon Isolation

Principles of the muon-isolation algorithms

4.1.4 High Level Triggering

4.2 Electrons and Photons

4.2.1 ECAL clustering and superclustering

superclustering algorithm

4.2.2 Electron and photon trigger

4.2.3 Photon reconstruction and selection

Photon energy reconstruction

Photon isolation

4.2.4 Electron reconstruction and selection

Track-cluster combination

Electron isolation

Chapter 5

Differential and integrated fiducial cross sections for Higgs boson production in the four-lepton decay channel

5.1 Introduction

In 2012, two general purpose detectors CMS and ATLAS announced the discovery of existence of new boson [7, 8] of mass around 125 GeV. The comprehensive properties has been studied by both CMS and ATLAS experiment [9, 10, 11, 12] of new discovered boson is found to be compatible with SM Higgs boson [1, 2, 3, 4, 5, 6].

In this chapter measurements of the inclusive and differential fiducial cross sections for the Higgs boson production in the $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) in proton collisions at center-of-mass energies of $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV are presented. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale. Any deviation of inclusive or differential distributions from SM expectations would provide a direct hint to BSM contributions.

The inclusive fiducial cross section measurements are performed using 5.1 fb⁻¹ of proton 7 TeV collision data and 19.7 fb⁻¹ of proton 8 TeV collision data recorded with the CMS detector at LHC. The measurement of the ratio of fiducial cross sections at 7 TeV and 8 TeV is also reported. The differential fiducial cross section measurements are measured using the 8 TeV collision data only, due to the limited statistics of the 7 TeV dataset. To minimize the dependance of the results on the underlying model of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial phase space closely matching with the experimental acceptance. The remaining model dependence is extracted by repeating the measurement for SM and other exotic model and quoted as a separate systematic uncertainty. All measurements include the correction for the finite

detection efficiency and resolution effects, and are performed assuming the Higgs boson mass of $m_H = 125.0$ GeV, as measured by the CMS experiment using the decay channels $H \rightarrow 4\ell$ and $H \rightarrow 2\gamma$ [52].

The differential fiducial cross section is measured as a function of kinematic properties of the four-lepton system and the leading jet, as well as a function of the jet multiplicity. The results are reported for several kinematic observables which are sensitive to the Higgs-boson production mechanism: transverse momentum and the rapidity of the four-lepton system, transverse momentum of the leading jet, as well as distance in rapidity between the Higgs boson candidate and the leading jet. The results are also reported for kinematic observables which are sensitive to the Higgs boson decay mechanism: the invariant masses of the two lepton pairs and five helicity angles formed by the leptons in the Collins-Soper frame [53].

The chapter is organized as follows. The datasets and simulated samples used in the analysis are described in Section 5.2. The analysis strategy is presented in Section 5.3, where we present the event selection, fiducial volume definition and procedures for performing the inclusive and differential fiducial cross sections measurements. In Section 5.4 we discuss the dominant systematic uncertainties considered in the measurements. In Section 5.5 we present the results of all the measurements and their comparison with the SM-based theoretical calculations. At end, summary is presented in Section 5.6.

5.2 Datasets, simulation samples and theoretical predictions

5.2.1 Datasets

The dataset used in the measurements was collected by the CMS experiment in 2011 and 2012, and corresponds to 5.1 fb^{-1} of 7 TeV collision data and 19.7 fb^{-1} of 8 TeV collision data respectively. All the details on the CMS standard selection of runs and luminosity sections, as well as the selection of the triggers used to collect this dataset are described in Ref. [54]. Thus, similar detector operation conditions are imposed for the validation of the data to be used for the analysis of the $4e$, 4μ and $2e2\mu$ channels. The absolute integrated luminosity of the proton collisions in 2011 has been determined with a relative precision of 2.2% [55], while in 2012 it has been determined with a relative precision of 2.6% [56]. The analysis relies on primary datasets (PDs) produced centrally and which combine various collections of High Level Triggers (HLT). The detailed content of the PDs evolves in phase with the evolution of the trigger menu to cope with ever increasing instantaneous luminosity. The analysis relies on the so-called *DoubleElectron* and *DoubleMuon*. These

latter PDs are formed by a OR between various triggers with symmetric or asymmetric trigger thresholds for the two leptons, with or without additional identification and isolation requirements. They also include triggers requiring three leptons above a low p_T threshold. It also includes cross-triggers which were added to recover few percent of inefficiency in the $2e2\mu$ channel at low Higgs boson masses, forming the so-called $MuEG$ PD. Tri-electron triggers for 2012 data are also used, in order to recover most of the inefficiency of DoubleElectron trigger. All the triggers used with the 8 TeV data collected in 2012 are listed in Table 5.1.

# Channel	Purpose	HLT path	L1 seed	prescale
4e	main	HLT_Ele17_CaloTrk_Ele8_CaloTrk	L1_DoubleEG_13_7	1
		OR HLT_Ele15_Ele8_Ele5_CaloIdL_TrkIdVL	L1_TripleEG_12_7_5	1
4μ	main	HLT_Mu17_Mu8	L1_Mu10_MuOpen	1
		OR HLT_Mu17_TkMu8	L1_Mu10_MuOpen	1
$2e2\mu$	main	HLT_Ele17_CaloTrk_Ele8_CaloTrk	L1_DoubleEG_13_7	1
		OR HLT_Mu17_Mu8	L1_Mu10_MuOpen	1
		OR HLT_Mu17_TkMu8	L1_Mu10_MuOpen	1
		OR HLT_Mu8_Ele17_CaloTrk	L1_MuOpen_EG12	1
		OR HLT_Mu17_Ele8_CaloTrk	L1_Mu12_EG6	1
4μ	backup	HLT_TripleMu5	L1_TripleMu0	1
4e and $2e2\mu$	Z T&P	HLT_Ele17_CaloTrkVT_Ele8_Mass50	L1_DoubleEG_13_7	5
4e and $2e2\mu$	Z T&P low p_T	HLT_Ele20_CaloTrkVT_SC4_Mass50_v1	L1_SingleIsoEG18er	10
4μ and $2e2\mu$	Z T&P	HLT_IsoMu24_eta2p1	L1_SingleMu16er	
4μ and $2e2\mu$	J/psi T&P	HLT_Mu7_Track7_Jpsi		
		HLT_Mu5_Track3p5_Jpsi		
		HLT_Mu5_Track2_Jpsi		

TABLE 5.1: triggers used with the 8 TeV data collected in 2012.

5.2.2 Simulation samples and theoretical predictions

Descriptions of the SM Higgs boson production in the gluon fusion ($gg \rightarrow H$) process are obtained using the HRES [57, 58], POWHEG [59, 60] and POWHEG MINLO-HJ [61] generators. The HRES generator is a partonic level Monte Carlo generator that provides a description of the $gg \rightarrow H$ process at next-to-next-to-leading order (NNLO) accuracy in perturbative QCD and next-to-next-to-leading logarithmic (NNLL) accuracy in resummation of soft-gluon effects at small transverse momenta [57, 58]. Since the resummation is inclusive over the QCD radiation recoiling against the Higgs boson, HRES is considered for the prediction of fiducial cross sections which are inclusive in the associated jet activity. The POWHEG generator is a partonic level matrix-element generator that implements next-to-leading (NLO) perturbative QCD calculations and additionally provides an interface to parton-shower programs. It provides a description of the $gg \rightarrow H$ production in association with zero jets at NLO accuracy. For the purpose of this analysis, it has been tuned to closely match the Higgs boson p_T spectrum in the full phase space as predicted by the HRES generator. The POWHEG MINLO-HJ generator is an extension

of the POWHEG generator based on the MiNLO prescription [61] for the improved next-to-leading logarithmic (NLL) accuracy applied to the $gg \rightarrow H$ production in association with up to one additional jet. It provides a description of the $gg \rightarrow H$ production in association with zero and one jets at the NLO accuracy, and the $gg \rightarrow H$ production in association with two jets only at the LO accuracy. Descriptions of the SM Higgs boson production in the vector boson fusion (VBF) process is obtained at NLO accuracy using also the POWHEG generator. The processes of SM Higgs boson production associated with gauge bosons (VH) or top-antitop quark pair ($t\bar{t}H$) are described at LO accuracy using PYTHIA 6.4 [62], and higher order diagrams are accounted for by PYTHIA's parton showering model. The Monte Carlo (MC) samples simulated with these generators are normalised to the inclusive SM Higgs boson production cross sections and branching fractions that correspond to the SM calculations at NNLO and NNLL accuracy, in accordance with the LHC Higgs Cross Section Working Group recommendations [63]. In order to estimate the dependance of the measurement procedure on the underlying assumption for the Higgs boson production mechanism, we have used the set of MC samples for individual production mechanisms described in the paragraph above. In addition, in order to estimate the dependance of the measurement on different assumptions of the Higgs boson properties, we have also simulated a range of samples that describe production and decay of exotic Higgs-like resonances to the four-lepton final state. These include spin-zero, spin-one and spin-two resonances with anomalous interactions with a pair of neutral gauge bosons ($ZZ, Z\gamma^*, \gamma^*\gamma^*$) described by higher-order operators, as discussed in details in Ref. [64]. The procedure used to estimate the model dependance effects will be presented in Section 5.4. All these samples are generated using the POWHEG generator for the description of NLO QCD effects in the production mechanism, and JHUGEN [65, 66, 67] to describe the decay of these exotic resonances to four leptons including all spin correlations. The list of Standard Model Higgs boson 8 TeV samples with corresponding values of cross sections can be found in Table 5.3, while all the signal Model Carlo samples studied in the analysis are listed in Table 5.2. A subset of these samples that is used to asses the model dependance of the measurement results is indicated in the table. In a similar way, the samples of the exotic spin-parity models and exotic production/decay modes are presented in Table ?? in Appendix ?. The MC event samples that are used to estimate contribution from the background process $gg \rightarrow$ were simulated using MCFM 6.7, while the background process $q\bar{q} \rightarrow 4\ell$ was simulated at NLO accuracy with POWHEG generator including s-, t- and u-channel diagrams. For the purpose of the $Z \rightarrow 4\ell$ cross-section measurements, we also have separately modelled contributions from the t- and u-channels of the $q\bar{q}(\rightarrow ZZ^*) \rightarrow 4\ell$ process at NLO accuracy with POWHEG generator. All event

TABLE 5.2: Standard Model signal Model Carlo samples studies in the analysis. Samples used for the estimation of the model dependance of the measurement results are denoted in the column “D.”.

Sample	Description	D.
SMHiggsToZZTo4L_M-125_8TeV-powheg15-JHUGenV3-pythia6	gg→H (POWHEG+JHU)	Y
SMHiggsToZZTo4L_M-126_8TeV-powheg15-JHUGenV3-pythia6	gg→H (POWHEG+JHU)	Y
GluGluToHToZZTo4L_M-125_8TeV-powheg15-pythia6	gg→H (POWHEG)	No
GluGluToHToZZTo4L_M-126_8TeV-powheg15-pythia6	gg→H (POWHEG)	No
GluGluToHToZZTo4L_M-125_8TeV-powheg-pythia6	gg→H (POWHEG1.0)	No
GluGluToHToZZTo4L_M-126_8TeV-powheg-pythia6	gg→H (POWHEG1.0)	No
GluGluToHToZZTo4L_M-125_8TeV-minloHJJ-pythia6-tauola	gg→H (MINLOHJJ)	Y
GluGluToHToZZTo4L_M-126_8TeV-minloHJJ-pythia6-tauola	gg→H (MINLOHJJ)	Y
VBF_HToZZTo4L_M-125_8TeV-powheg-pythia6	VBF (POWHEG)	Y
VBF_HToZZTo4L_M-126_8TeV-powheg-pythia6	VBF (POWHEG)	Y
WH_HToZZTo4L_M-125_8TeV-pythia6	WH (PYTHIA)	Y
WH_HToZZTo4L_M-126_8TeV-pythia6	WH (PYTHIA)	Y
ZH_HToZZTo4L_M-125_8TeV-pythia6	ZH (PYTHIA)	Y
ZH_HToZZTo4L_M-126_8TeV-pythia6	ZH (PYTHIA)	Y
TTbarH_HToZZTo4L_M-125_8TeV-pythia6	ttH (PYTHIA)	Y
TTbarH_HToZZTo4L_M-126_8TeV-pythia6	ttH (PYTHIA)	Y

generators described above take into account the internal initial-state and final-state radiation effects which can lead to the presence of additional hard photons in an event. The POWHEG and JHUGEN event generators also properly take into account interference between all contributing amplitudes, including those related to the permutations of identical leptons in the $4e$ and 4μ final states. In case of the LO generators, the set of parton distribution functions (PDF) CTEQ6L [68] is used, in case of the NLO generators the PDF set CT10 [69] is used, while the PDF set MSTW2008 [70] is used in case of the NNLO generator. All generated events are interfaced to PYTHIA 6.4 Tune Z2* [71] to simulate the effects of the parton shower, hadronization and underlying event. The PYTHIA 6.4 Z2* tune is derived from the Z1 tune [72], which uses the CTEQ5L parton distribution set, whereas Z2* adopts CTEQ6L [73]. In case of HRES, since it does not provide an interface to programs that can simulate the hadronization and underlying event effects, the size of these effects is first estimated using the POWHEG+JHUGEN and its interface to PYTHIA 6.4, and then the correction is applied to the HRES parton level predictions. The generated events were processed through a detailed simulation of the CMS detector based on [74] and were reconstructed with the same algorithms

that are used for data analysis. Multiple interactions (pileup) are included in simulations to match the distribution of the number of interactions per LHC bunch crossing observed in data. The average number of pileup interactions is estimated to be approximately 9 and 21 in the 7 and 8 TeV datasets, respectively. Energy deposits from pileup interactions and from the underlying event are subtracted when computing the energy of jets and isolation of reconstructed objects using the FASTJET technique [75, 76, 77]. This technique is based on the calculation of the η -dependent average energy density evaluated individually for each event.

The selection efficiency in all simulated samples is rescaled in order to correct for residual differences in lepton selection efficiencies in data and simulation. This correction is based on the total lepton selection efficiencies measured in inclusive samples of Z-boson events in simulation and data using a “tag and probe” method [78], separately for 7 and 8 TeV collisions. The dependency of the selection efficiency on the number of reconstructed primary vertices in the event is negligible for both the 7 and 8 TeV data samples. More details are presented and discussed in Ref. [54].

Process	MC	$\sigma_{(N)NLO}$	Comment
$gg \rightarrow H(\rightarrow ZZ) \rightarrow 4\ell$	POWHEG+JHUGEN	19.27 pb	$gg \rightarrow H$ production
$qq \rightarrow H(\rightarrow ZZ) + qq \rightarrow 4\ell + X$	POWHEG	1.578 pb	VBF production
$qq \rightarrow H(\rightarrow ZZ) + W \rightarrow 4\ell + X$	PYTHIA	0.7046 pb	WH production
$qq \rightarrow H(\rightarrow ZZ) + Z \rightarrow 4\ell + X$	PYTHIA	0.4153 pb	ZH production
$qq \rightarrow H(\rightarrow ZZ) + t\bar{t} \rightarrow 4\ell + X$	PYTHIA	0.1293 pb	ttH production

TABLE 5.3: MC simulation datasets used for the signal processes predicted by the SM for $m(H)=125.0$ GeV (Z stands for Z, Z^* , γ^*). The cross sections do not include the ZZ and 4ℓ branching ratios. In the SM, the branching ratio $H \rightarrow ZZ \rightarrow 4\ell$ is 1.247×10^{-4} for $m(H)=125.0$ GeV.

5.3 Analysis strategy

The inclusive and differential and inclusive cross sections are measured in a fiducial region in order to reduce the effects of model dependent acceptances. The fiducial selection mimics the reconstruction level selection, which is optimised to detect a low mass ($m \sim 125$ GeV) Higgs boson decaying to 4 leptons through two Z bosons. Since in the standard model there are multiple production modes, the fiducial selection and measurement strategy are designed to be independent of

Process	MC	$\sigma_{(N)NLO}$	Comment
$q\bar{q} \rightarrow ZZ \rightarrow 4e(4\mu, 4\tau)$	POWHEG	76.91 fb	s+t/u channels
$q\bar{q} \rightarrow ZZ \rightarrow 2e2\mu$	POWHEG	176.7 fb	s+t/u channels
$q\bar{q} \rightarrow ZZ \rightarrow 2e(2\mu)2\tau$	POWHEG	176.7 fb	s+t/u channels
$q\bar{q} \rightarrow ZZ \rightarrow 4e$	POWHEG	494.8 fb	t/u channels only, $m_{ll} > 1\text{GeV}$
$q\bar{q} \rightarrow ZZ \rightarrow 2e2\mu$	POWHEG	1527 fb	t/u channels only, $m_{ll} > 1\text{GeV}$
$q\bar{q} \rightarrow ZZ \rightarrow 4\mu$	POWHEG	488.4 fb	t/u channels only, $m_{ll} > 1\text{GeV}$
$gg \rightarrow ZZ \rightarrow 2e2\mu$	MCFM 6.7	2.444 fb	K-factor of 2.66 included
$gg \rightarrow ZZ \rightarrow 4\mu$	MCFM 6.7	1.222 fb	K-factor of 2.66 included
$gg \rightarrow ZZ \rightarrow 4e$	MCFM 6.7	1.222 fb	K-factor of 2.66 included

TABLE 5.4: MC simulation datasets used for the background processes (Z stands for Z, Z^*, γ^*); In MC samples with label “t/u channels only” only the t/u channels are simulated in the $qq \rightarrow 4\ell$ process and these samples are used for the $Z \rightarrow 4\ell$ analysis. The “t/u channels only” samples have lower cut in m_{ll} of 1 GeV, while all other $qq \rightarrow ZZ$ and $gg \rightarrow ZZ$ samples have lower cut in m_{ll} of 4 GeV. In case of the $gg \rightarrow ZZ$ background, the overall NLO k-factor of 2.66 is included, as discussed in Section 5.3.1.

how the Higgs boson is produced. For this reason, the inclusion of isolation in the fiducial selection is necessary in order to make the reconstruction efficiency with respect to the fiducial volume independent of the number of jets.

Furthermore, in the sub-leading SM production modes there can be associated vector bosons which can decay to leptons, and therefore the reconstructed Higgs boson candidate may not be composed of the correct combination of leptons. In order to retain the full information from an unbinned simultaneous fit of signal and background pdfs to the four-lepton mass distribution, it is necessary to treat such “wrong combination” events as background, i.e. not within the fiducial volume.

Further signal-induced background can be formed due to imperfect correspondence between quantities at the particle level and the corresponding quantities at the reconstruction level. Signal events which are from within the fiducial volume but fail the reconstruction are accounted for in the fiducial efficiency (ϵ) and the expected contribution from events which pass the reconstruction selection but are not from within the fiducial volume (f_{out}) are referred to as “out-of-acceptance” events and are treated as background.

The effects of imperfect detector resolution can in general have a non-negligible impact on the shape of the distribution of the measured observables. For this reason, in case of the differential cross sections measurements, a procedure to correct

for the detection efficiencies and resolution effects is applied. Throughout this document, this procedure will be referred to as the unfolding procedure, and the unfolded differential distributions will be referred to as distributions at the fiducial level. The reconstruction level selection, fiducial volume definition, statistical procedure, and their model dependence will be described the subsequent sections.

5.3.1 Event selection and background modeling

Event selection

The events selection for this measurement is similar to events selection used in previous Higgs boson properties measurement in the same final state [54, 64]. The online event selection is based on single and double lepton triggers, as well as triple electron triggers. The minimum p_T of 17 and 8 GeV is required for leading and subleading leptons for double leptons triggers, while 15, 8 and 5 GeV for triple electron trigger. In offline selection, events are selected with atleast four well identified and isolated leptons produced at primary vertex. Z_1 is reconstructed with invariant mass closest to the world average Z boson mass by all possible combination of same flavor and opposite charge (SFOS) pairs of leptons within mass range of $40 \leq m(Z_1) \leq 120$ GeV. Z_2 is reconstructed by remaining SFOS pair of leptons within inv. mass range of $12 \leq m(Z_2) \leq 120$ GeV. If more than one Z_2 passes the criteria, the leptons pair with largest $\Sigma|p_T|$ is selected. There should be atleast two leptons with $p_T \geq 20$ and ≥ 10 among four leptons selected to form Z_1 and Z_2 . To remove contamination from QCD low masses resonance, any opposite-charge pair of leptons is required to pass $m(\ell^+\ell'^-) \geq 4$ GeV. The final state radiated (FSR) photons are also recovered with algorithm described in [54].

Background modeling

The dominant and sub-dominant irreducible backgrounds for $H \rightarrow 4\ell$ process are ZZ production via $q\bar{q}$ annihilation and gg fusion respectively. The reducible background is off instrumental nature arises from the process where partial jets are identified as leptons. The main process which contributes in this background are: jets production in association with Z boson, jets production in association with WZ boson pair and the $t\bar{t}$ pair production. From now on, we would be represented this background as .

In case of the the inclusive and differential $H \rightarrow 4\ell$ measurements, the irreducible $q\bar{q} \rightarrow ZZ$ and $gg \rightarrow ZZ$ backgrounds are evaluated from Monte Carlo simulation, following Ref. [54]. In addition, following the study in Ref. [79], in



these measurements we have increased the expected yield for the $gg \rightarrow ZZ$ background by assigning to the leading order (LO) background cross section a $m_{4\ell}$ dependent k-factor equal to the one used for the signal $H \rightarrow 4\ell$ process. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of the expected $gg \rightarrow ZZ$ background by a factor of ~ 2.66 , with the systematic uncertainty of 10% (to be changed to 60%). After the inclusion of this k-factor, the $gg \rightarrow ZZ$ process is still a subdominant background in all $H \rightarrow 4\ell$ measurements.

The reducible background (Z+X) is evaluated using a tight-to-loose “fake rate” method and the control regions in the data, following Ref. [54]. In case of the inclusive $H \rightarrow 4\ell$ fiducial cross-section measurement, the shape of the $m_{4\ell}$ distribution for the reducible background is obtained by fitting the $m_{4\ell}$ with empirical analytical functional forms (ref. [54]). Prediction for the reducible background in all three channels together fitted using this empirical shape with indicated total uncertainty is shown in Figure 5.2. In case of the differential $H \rightarrow 4\ell$ measurements, the shape of the reducible background is obtained from the templates built from the control

TABLE 5.5: The number of estimated background and signal events, as well as number of observed candidates, after final inclusive selection in the range $105 < m_{4\ell} < 140$ GeV used in the $H \rightarrow 4\ell$ measurements. Signal and ZZ background are estimated from simulations, while Z+X background is estimated from control regions in data.

Channel	4e	4 μ	2e2 μ
19.7 fb ⁻¹ @ 8 TeV			
$q\bar{q} \rightarrow ZZ$	3.05 ± 0.35	7.62 ± 0.49	9.03 ± 0.69
Z + X	1.52 ± 0.34	1.18 ± 0.48	4.16 ± 1.06
$gg \rightarrow ZZ$	0.20 ± 0.05	0.41 ± 0.10	0.50 ± 0.13
Total background expected	4.77 ± 0.71	9.21 ± 0.69	13.69 ± 1.29
$H \rightarrow 4l$ ($m_H = 125.0$ GeV)	2.93 ± 0.45	5.63 ± 0.67	7.29 ± 0.93
Observed	9	15	15
5.1 fb ⁻¹ @ 7 TeV			
$q\bar{q} \rightarrow ZZ$	0.84 ± 0.10	1.80 ± 0.11	2.24 ± 0.28
Z + X	0.34 ± 0.09	0.22 ± 0.09	1.03 ± 0.29
$gg \rightarrow ZZ$	0.03 ± 0.01	0.06 ± 0.02	0.07 ± 0.02
Total background expected	1.21 ± 0.14	2.08 ± 0.14	3.37 ± 0.40
$H \rightarrow 4l$ ($m_H = 125.0$ GeV)	0.70 ± 0.11	1.24 ± 0.14	1.67 ± 0.26
Observed	1	3	6

regions in data for each bin of the observable considered for the measurement, following a procedure similar to the one used in the spin-parity studies described in the Ref. [54]. More details on this procedure and template shapes for each of the differential $H \rightarrow 4\ell$ measurements will be given in Section 5.3.5.

The number of estimated background and signal events, as well as the number of observed candidates after final inclusive selection in data in the narrow mass region around 125.0 GeV used in the $H \rightarrow 4\ell$ measurements ($105 < m_{4\ell} < 140$ GeV) are given in Table 5.5, separately for 7 TeV and 8 TeV. These numbers are presented and discussed in detail in Ref. [54].

The reconstructed four-lepton invariant mass distribution in the region of interest for the $H \rightarrow 4\ell$ and $Z \rightarrow 4\ell$ measurements described in this document ($50 < m_{4\ell} < 140$ GeV) is shown in the Figure 5.1 for the full 7 TeV and 8 TeV datasets, and compared to the SM expectations. Prediction for the reducible background in a wide range of $m_{4\ell}$ in all three channels together, fitted using the empirical shape with indicated total uncertainty, is separately shown in Figure 5.2.

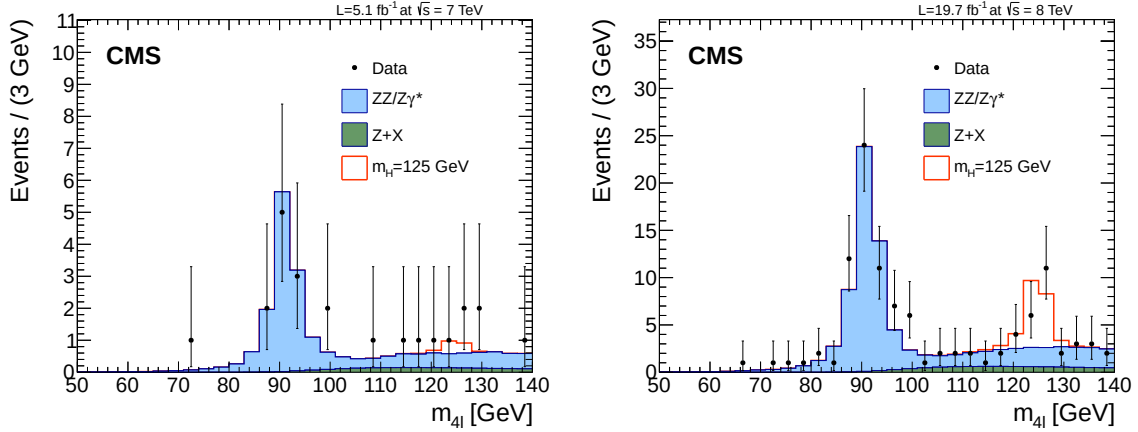


FIGURE 5.1: Distributions of the $m_{4\ell}$ observable in 7 TeV (left) and 8 TeV (right) datasets, as well as expectations for the SM Higgs boson ($m_H = 125.0$ GeV) and contributing SM processes, including the resonant $Z \rightarrow 4\ell$ decays.

5.3.2 Definition of the fiducial volume

The fraction of all $H \rightarrow 4\ell$ decays that get selected following the event selection procedure above can vary significantly between different Higgs boson production mechanisms and different exotic models of Higgs boson properties. In processes with large jet activity (such as $t\bar{t}H$ production) or with the $H \rightarrow 4\ell$ kinematics

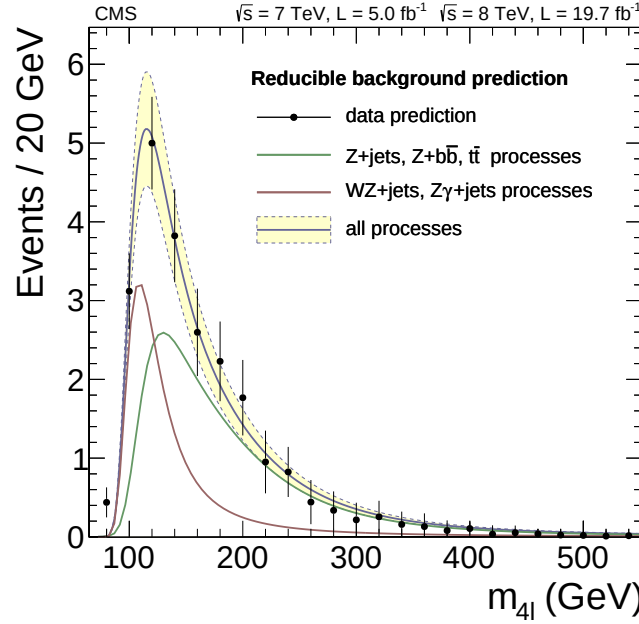


FIGURE 5.2: Prediction for the reducible background in all three channels together (black dots) fitted using an empirical shape (blue curve) with indicated total uncertainty (yellow band). Contributions from the 2P2F-like (green) and 3P1F-like (red) processes are fitted separately.

different from the SM prediction (such as $H \rightarrow Z\gamma^*(\gamma^*\gamma^*) \rightarrow 4\ell$ processes, and exotic Higgs-like spin-one models) the acceptance of signal events can be up to 70% different from the acceptance predicted for the SM $H \rightarrow 4\ell$ decays. In order to minimise the dependence of the measurement results on the underlying assumption on model of the Higgs boson production and properties, the fiducial phase space for the $H \rightarrow 4\ell$ cross section measurements is defined to match as closely as possible the experimental acceptance defined by the reconstruction level selection.

The fiducial volume requires at least four leptons (electrons, muons), with at least one lepton having $p_T > 20$ GeV, an another lepton having $p_T > 10$ GeV, and the remaining electrons (muons) having $p_T > 7(5)$ GeV. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code = 3), before any final state radiation occurs. All electrons (muons) must have pseudorapidity $|\eta| < 2.5(2.4)$. In addition, each lepton must satisfy an isolation requirement computed using the p_T sum of all stable particles (PYTHIA status code = 1) within $\Delta R < 0.4$ distance from that lepton. The p_T sum excludes any neutrinos, as well

as any photon or stable lepton which is a daughter of the lepton for which the isolation sum is being computed. The ratio of this sum and the p_T of the considered lepton must be less than 0.4, in line with the requirement on the lepton isolation at the reconstruction level [54]. Inclusion of isolation is an important step in the fiducial volume definition as it reduces significantly the differences in signal selection efficiency between different models. It has been shown in simulation that signal selection efficiency differs up to 45% between different models if lepton isolation requirement is not included.

Furthermore, several topological selections are applied as part of the fiducial phase space definition. There must exist at least two SFOS lepton pairs, and all SFOS lepton pairs are used to form Z-boson candidates. The SFOS pair with invariant mass closest to the nominal Z-boson mass (91.2 GeV) is taken as the first Z-boson candidate (denoted as Z_1). The mass of the Z_1 candidate must satisfy $40 \text{ GeV} < m(Z_1) < 120 \text{ GeV}$. The remaining set of SFOS pairs are used to form the second Z-boson candidate (denoted as Z_2). In events with more than one Z_2 candidate, the SFOS pair with the highest $\Sigma|p_T|$ is chosen. The mass of the Z_2 candidate must satisfy $12 \text{ GeV} < m(Z_2) < 120 \text{ GeV}$. The four selected leptons must satisfy $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. Of the four selected leptons, the invariant mass of any opposite sign lepton pair must satisfy $m(\ell^+ \ell'^-) > 4 \text{ GeV}$. Finally, all four selected leptons must originate from the decay of the Higgs boson, and the invariant mass of the Higgs-boson candidate must satisfy $105 \text{ GeV} < m(4\ell) < 140 \text{ GeV}$. The requirement on the $m(4\ell)$ is important as the off-shell production cross section in the dominant gluon fusion production mode is sizeable and can amount up to a few percent of the total cross-section [80, 81].

It has been verified in simulation that the reconstruction efficiency for events originating from the fiducial phase space defined in this way only weakly depends on the Higgs boson production mechanism and its properties. The systematic effect associated with the remaining model dependance will be extracted and quoted separately, as described in Section 5.4. Fraction of signal events within the fiducial volume $\mathcal{A}_{\text{fid}}$, and reconstruction efficiency ϵ for signal events from within the fiducial volume for individual SM production modes and exotic signal models are listed in Table 5.6 and 5.7. In case of WH, ZH and $t\bar{t}H$ production modes the event selection procedure will occasionally select 4ℓ candidates where at least one reconstructed lepton originates from associated vector bosons or jets, and not from the $H \rightarrow 4\ell$ decays. Fraction of such signal events in the mass range $105 \text{ GeV} < m(4\ell) < 140 \text{ GeV}$ amounts to about 4%, 18% and 22% for WH, ZH and $t\bar{t}H$ production modes, respectively. These events have a broad invariant $m_{4\ell}$ mass distribution that depends on the exact production mechanism. The fiducial phase space definition formulated above does not include this type of events, and these events will be treated as a non-resonant signal-induced background in the



TABLE 5.6: The fraction of signal events within the fiducial volume (acceptance $\mathcal{A}_{\text{fid}}$), reconstruction efficiency (ϵ) for signal events from within the fiducial volume, and ratio of reconstructed events which are from outside the fiducial volume to reconstructed events which are from within the fiducial volume (f_{nonfid}). Values are given for characteristic 8 TeV signal models assuming $m_H = 125.0$ GeV, while uncertainties include statistical uncertainties due to limited number of events in MC simulation.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
$gg \rightarrow H$ (POWHEG+JHUGEN)	0.422 ± 0.001	0.647 ± 0.002	0.053 ± 0.001	0.681 ± 0.002
$gg \rightarrow H$ (MINLOHJJ)	0.415 ± 0.001	0.636 ± 0.002	0.051 ± 0.001	0.668 ± 0.002
VBF (POWHEG)	0.476 ± 0.003	0.652 ± 0.005	0.040 ± 0.002	0.678 ± 0.005
WH (PYTHIA)	0.342 ± 0.002	0.627 ± 0.003	0.072 ± 0.002	0.672 ± 0.003
ZH (PYTHIA)	0.348 ± 0.003	0.634 ± 0.004	0.072 ± 0.003	0.679 ± 0.005
$t\bar{t}H$ (PYTHIA)	0.250 ± 0.003	0.601 ± 0.008	0.139 ± 0.008	0.685 ± 0.010
Some characteristic models of Higgs-like boson with exotic decays and properties				
$q\bar{q} \rightarrow H(J^{\text{CP}}=1^-)$ (JHUGEN)	0.238 ± 0.001	0.609 ± 0.002	0.054 ± 0.001	0.642 ± 0.002
$q\bar{q} \rightarrow H(J^{\text{CP}}=1^+)$ (JHUGEN)	0.283 ± 0.001	0.619 ± 0.002	0.051 ± 0.001	0.651 ± 0.002
$gg \rightarrow H \rightarrow Z\gamma^*$ (JHUGEN)	0.156 ± 0.001	0.622 ± 0.002	0.073 ± 0.001	0.667 ± 0.002
$gg \rightarrow H \rightarrow \gamma^*\gamma^*$ (JHUGEN)	0.188 ± 0.001	0.629 ± 0.002	0.066 ± 0.001	0.671 ± 0.002

measurement procedure. This approach provides a simple measurement procedure with a reduced signal model dependence. More details will be discussed in Section 5.3.

5.3.3 Signal model and statistical procedure

The differential cross section within the fiducial volume is extracted in bins of the kinematic observables of interest and corrected for the detection efficiency and resolution effects. The inclusive cross section measurement is treated as a special case where the number of bins is one.

In general, it is not enough to build the signal model for each bin of the observables measured at the reconstruction level. To unfold the reconstruction-level distribution to the distribution before all detector effects, the efficiency matrix ϵ is constructed using the signal MC samples. The efficiency matrix ϵ provides the probability of an event that was in i -th bin at the fiducial level and is reconstructed

TABLE 5.7: The fraction of signal events within the fiducial volume (acceptance $\mathcal{A}_{\text{fid}}$), reconstruction efficiency (ϵ) for signal events from within the fiducial volume, and ratio of reconstructed events which are from outside the fiducial volume to reconstructed events which are from within the fiducial volume (f_{nonfid}). Values are given for characteristic 8 TeV signal models assuming $m_H = 125.0$ GeV, while uncertainties include statistical uncertainties due to limited number of events in MC simulation.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
$gg \rightarrow H$ (POWHEG+JHUGEN)	0.438 ± 0.001	0.676 ± 0.002	0.049 ± 0.001	0.709 ± 0.002
$gg \rightarrow H$ (MINLOHJJ)	0.427 ± 0.001	0.675 ± 0.002	0.047 ± 0.001	0.707 ± 0.002
VBF(POWHEG)	0.483 ± 0.003	0.674 ± 0.004	0.035 ± 0.002	0.697 ± 0.005
WH(PYTHIA)	0.357 ± 0.002	0.660 ± 0.004	0.061 ± 0.002	0.700 ± 0.050
ZH(PYTHIA)	0.372 ± 0.004	0.662 ± 0.006	0.059 ± 0.004	0.701 ± 0.007
$t\bar{t}H$ (PYTHIA)	0.248 ± 0.004	0.636 ± 0.010	0.158 ± 0.011	0.736 ± 0.013
Some characteristic models of Higgs-like boson with exotic decays and properties				
$q\bar{q} \rightarrow H(J^P = 1^-)$ (JHUGEN)	0.254 ± 0.001	0.645 ± 0.002	0.049 ± 0.001	0.677 ± 0.002
$q\bar{q} \rightarrow H(J^P = 1^+)$ (JHUGEN)	0.301 ± 0.001	0.657 ± 0.002	0.044 ± 0.001	0.686 ± 0.002
$gg \rightarrow H \rightarrow Z\gamma^*$ (JHUGEN)	0.160 ± 0.001	0.654 ± 0.002	0.072 ± 0.001	0.701 ± 0.002
$gg \rightarrow H \rightarrow \gamma^*\gamma^* (J^P = 0^-)$ (JHUGEN)	0.185 ± 0.001	0.659 ± 0.002	0.064 ± 0.001	0.701 ± 0.002

in j -th bin as:

$$\epsilon_{i,j} = \frac{N(j_{\text{reco}} \&\& i_{\text{gen}})}{N(i_{\text{gen}})}, \quad (5.1)$$

where $N(i_{\text{gen}})$ is the number of event in i -th bin at the fiducial level and $N(j_{\text{reco}} \&\& i_{\text{gen}})$ is the number of event that was in i -th bin at the fiducial level and is reconstructed in j -th bin. Fig. ?? shows the efficiency matrix for $p_T(H)$ measurement using different SM production MC samples. The efficiency matrix is dominated by diagonal elements, but in general it can have non-negligible off-diagonal elements. As a result, the number of signal events within the fiducial volume in j -th bin for the reconstructed observables can be parametrized as:

$$S_{\text{fid}}(j_{\text{reco}}) = \sum_{i_{\text{gen}}} L \times \epsilon_{i,j} \times \sigma_{i_{\text{gen}}}, \quad (5.2)$$

where L is the luminosity and $\sigma_{i_{\text{gen}}}$ is the fiducial cross section in i -th bin of kinematic observables at the fiducial level.

TABLE 5.8: Fiducial volume acceptance per final state for different Standard Model signal models.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG+JHUGEN) 125 GeV	0.409 ± 0.003	0.453 ± 0.003	0.413 ± 0.002	0.422 ± 0.001
gg→H (POWHEG+JHUGEN)126 GeV	0.410 ± 0.003	0.460 ± 0.003	0.422 ± 0.002	0.429 ± 0.001
gg→H (POWHEG) 125 GeV	0.396 ± 0.003	0.437 ± 0.003	0.411 ± 0.002	0.413 ± 0.001
gg→H (POWHEG) 126 GeV	0.402 ± 0.003	0.447 ± 0.003	0.418 ± 0.002	0.421 ± 0.001
gg→H (POWHEG1.0) 125 GeV	0.395 ± 0.001	0.443 ± 0.001	0.411 ± 0.001	0.415 ± 0.001
gg→H (POWHEG1.0) 126 GeV	0.408 ± 0.003	0.446 ± 0.003	0.421 ± 0.002	0.424 ± 0.001
gg→H (MINLOHJJ) 125 GeV	0.394 ± 0.002	0.439 ± 0.002	0.413 ± 0.002	0.415 ± 0.001
gg→H (MINLOHJJ) 126 GeV	0.407 ± 0.002	0.449 ± 0.002	0.418 ± 0.002	0.423 ± 0.001
VBF (POWHEG) 125 GeV	0.456 ± 0.007	0.503 ± 0.007	0.473 ± 0.005	0.476 ± 0.003
VBF (POWHEG) 126 GeV	0.450 ± 0.007	0.513 ± 0.007	0.471 ± 0.005	0.476 ± 0.003
WH (PYTHIA) 125 GeV	0.332 ± 0.003	0.365 ± 0.003	0.336 ± 0.002	0.342 ± 0.002
WH (PYTHIA) 126 GeV	0.331 ± 0.003	0.365 ± 0.003	0.340 ± 0.002	0.344 ± 0.002
ZH (PYTHIA) 125 GeV	0.343 ± 0.005	0.367 ± 0.005	0.341 ± 0.004	0.348 ± 0.003
ZH (PYTHIA) 126 GeV	0.339 ± 0.005	0.384 ± 0.005	0.362 ± 0.004	0.362 ± 0.003
ttH (PYTHIA)125 GeV	0.239 ± 0.007	0.274 ± 0.007	0.244 ± 0.005	0.250 ± 0.003
ttH (PYTHIA)126 GeV	0.261 ± 0.007	0.271 ± 0.007	0.255 ± 0.005	0.260 ± 0.004

The normalisation of the reconstructed events that fall out of the fiducial volume, identified as out-of-acceptance events, are related to the normalisation of the normalisation of events within the fiducial volume by the out-of-acceptance fraction f_{out} for j-th reco bin:

$$f_{\text{out}}^j = \frac{N(\text{j}_{\text{reco}} \&\& \text{not fid.})}{N(\text{j}_{\text{reco}} \&\& \text{fid.})}. \quad (5.3)$$

Finally, the unfolding procedure to extract the fiducial cross section in bins of kinematic observables at the fiducial level, corrected for the detection efficiency and resolution effects, is performed with a maximum likelihood technique using the likelihood function:

$$\mathcal{L} = -2 \sum_{\text{j}_{\text{reco}}} \text{Log}(\mathcal{L}(S_{\text{in}} + S_{\text{out}} + S_{\text{wrong}} + B | N_{\text{j}_{\text{reco}}})) . \quad (5.4)$$

Here B represents the background expectation in each bin at the reconstruction level, whose mass spectrum, in general, can have non-negligible correlation with the observables. More details on the background mass spectra pdf($m_{4\ell}|\text{bkg}$) will be given in Section ??.

The likelihood function for j -th reco bin can be expanded as

$$\begin{aligned}
L(j_{\text{reco}}) &= C(j_{\text{reco}}) \cdot (N_{\text{sig}}^{\text{ch}}(m_{4\ell}) + N_{\text{bkg}}^{\text{ch}}(m_{4\ell})) \\
&= C(j_{\text{reco}}) \cdot (N_{\text{fid}}^{\text{ch}}(m_{4\ell}) + N_{\text{wrong}}^{\text{ch}}(m_{4\ell}) + N_{\text{out}}^{\text{ch}}(m_{4\ell}) + N_{\text{bkg}}^{\text{ch}}(m_{4\ell})) \\
&= C(j_{\text{reco}}) \cdot \left\{ \left(\sum_{i_{\text{gen}}} \sigma_{i_{\text{gen}}} \cdot \epsilon_{ij} \right) \cdot L \cdot (1 + f_{\text{out}}) \cdot \text{pdf}(m_{4\ell} | H \rightarrow 4\ell) \right. \\
&\quad \left. + N_{\text{wrong}}^{\text{ch}}(\text{reco}, j) \cdot \text{pdf}(m_{4\ell} | H \rightarrow \text{other}) \right. \\
&\quad \left. + N_{\text{bkg}}^{\text{ch}}(\text{reco}, j) \cdot \text{pdf}(m_{4\ell} | \text{bkg}) \right\}, \tag{5.5}
\end{aligned}$$

where $C(j_{\text{reco}})$ is a normalization factor.

The maximizing likelihood is performed by fitting simultaneously the signal cross section in each bin at the fiducial level with all reconstruction bin and all three final states. The exact details how the procedure outlined here is applied in case of the $H \rightarrow 4\ell$ inclusive and differential fiducial cross-section measurements will be given in the following Section ?? and Section ??.

5.3.4 Extraction of inclusive $H \rightarrow 4\ell$ fiducial cross section

5.3.5 Extraction of differential $H \rightarrow 4\ell$ fiducial cross sections

5.4 Systematic uncertainties

5.5 Results

5.5.1 Measurement of inclusive fiducial cross section

5.5.2 Measurement of differential fiducial cross sections

5.6 Summary

Chapter 6

Differential and integrated fiducial cross sections for Z boson production in the four-lepton decay channel

6.1 Extraction of $Z \rightarrow 4\ell$ fiducial cross sections

6.2 Extraction of ratio of $Z \rightarrow 4\ell$ and $H \rightarrow 4\ell$

6.3 Results

6.4 Summary

Chapter 7

Higgs boson production cross sections dependence on centre-of-mass energy

7.1 introduction

7.2 Higgs fiducial cross section measurements vs. $\sqrt{s}$

7.3 Summary

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